# Point convergence and curvature selection in smooth convex Nesterov flows

**Abstract**

We study point convergence and curvature selection for the equation $X'' + \alpha X'/t + \nabla F(X) = 0$. In finite dimensions, universal convergence from regular zero-velocity data at the singular time holds exactly for $\alpha \geq 3$, even when failure below the threshold is sought among globally smooth convex objectives with every positive-order derivative bounded. For each fixed $1 < \alpha < 3$, the construction can be refined so that the bounded trajectory has omega-limit equal to the entire compact minimizer segment. A reflected harmonic estimate yields this full-segment refinement. We then study confined valleys with one tangential and one normal coordinate. Exact quadratic valleys admit a positive limiting rescaled normal action and selection of minimum transverse curvature for $1 < \alpha \leq 2$. This conclusion persists for analytic higher normal terms under explicit compact confinement. Exact homogeneous valleys of order $2m$, $m \geq 2$, admit energy decay without action preparation, while finite-time prepared data yield positive action and coefficient selection for $1 < \alpha \leq (m+1)/m$. Actual confined nonselecting trajectories show the boundary of that selection range. The local valley conclusions retain their confinement and initialization hypotheses; they do not settle point convergence for general semialgebraic or other tame convex objectives.

**Keywords:** Nesterov flow; convex optimization; point convergence; omega-limit set; vanishing damping; adiabatic action; curvature selection

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# 1 The equation and the global threshold

Let $F : \mathbb{R}^d \to \mathbb{R}$ be differentiable and convex, with globally Lipschitz gradient and a nonempty minimizer set. We consider

$$X''(t) + \frac{\alpha}{t} X'(t) + \nabla F(X(t)) = 0 \quad (t > 0), \qquad X(0) = x_0, \qquad X'(0) = 0, \tag{1.1}$$

where $\alpha > 0$. A *regular cold solution* is a function $X \in C^2([0,\infty); \mathbb{R}^d)$ satisfying these conditions. The word cold always refers here to the singular time $t = 0$. A trajectory prescribed at a positive time will be described separately.

Subtracting the minimum, we take $F \geq 0$. The mechanical energy and the omega-limit set are

$$E(t) = \tfrac{1}{2}|X'(t)|^2 + F(X(t)), \qquad \omega(X) = \{z : \ X(t_j) \to z \text{ for some } t_j \to \infty\}. \tag{1.2}$$

Energy satisfies $E' = -\alpha |X'|^2/t$. Energy convergence and convergence to one point are different questions when the minimum is nonisolated.

**Theorem 1.1** (Universal damping threshold)**.** *Among positive damping parameters, the following property holds exactly for $\alpha \geq 3$: every regular cold solution of* (1.1)*, for every finite dimension and every differentiable convex objective with globally Lipschitz gradient and a nonempty minimizer set, converges to a minimizer. The same threshold holds in the smaller class of globally $C^\infty$ convex objectives satisfying*

$$\|D^j F\|_{L^\infty(\mathbb{R}^d)} < \infty \qquad (j \geq 1). \tag{1.3}$$

*For every fixed $1 < \alpha < 3$, failure occurs in dimension two with a bounded cold trajectory and a compact minimizer segment. The objective can be chosen so that*

$$X(0) = (0, 1), \qquad \omega(X) = \operatorname{argmin} F = [-S, S] \times \{0\} \tag{1.4}$$

*for some $S > 0$. For $0 < \alpha \leq 1$, a one-dimensional smooth example has an unbounded cold trajectory and a minimizer half-line.*

The objective in (1.4) may depend on the fixed parameter $\alpha$. Neither a common objective for a whole interval of damping parameters nor a discrete-time counterexample is asserted.

The existence of the smooth bounded counterexample with two distinct minimizing cluster points is proved in Appendix A. Section 2 establishes (1.4) from that complete construction. We give the endpoint arguments here, so that the quantifiers of the threshold are visible independently of the construction.

## 1.1 Regularity at the singular time

For a globally Lipschitz force $G$, the cold equation is equivalent to

$$X'(t) = -t^{-\alpha} \int_0^t v^\alpha G(X(v))\,\mathrm{d}v, \qquad X(t) = x_0 - \int_0^t \int_0^r (r^{-\alpha} v^\alpha) G(X(v))\,\mathrm{d}v\,\mathrm{d}r. \tag{1.5}$$

The position operator has Lipschitz constant at most $LT^2/[2(\alpha+1)]$ on $[0, T]$. It is therefore a contraction on a suitable closed ball for small $T$. Differentiation gives $X'(t) = -tG(x_0)/(\alpha+1) + o(t)$ and

$$X''(0) = -\frac{G(x_0)}{\alpha + 1}.$$

The equation then gives continuity of $X''$ at zero. Conversely, integrating the equation for $t^\alpha X'$ yields (1.5). Positive-time continuation is global because a globally Lipschitz force has at most linear growth. This argument works for every $\alpha > 0$, including $\alpha = 1$.

## 1.2 Convergence at and above three

**Proposition 1.2.** *Every regular cold solution in Theorem 1.1 converges when $\alpha \geq 3$.*

**Proof.** Put $a = \alpha - 1$ and fix a minimizer $z$. Define

$$\mathcal{E}_z(t) = t^2 F(X(t)) + \tfrac{1}{2}|tX'(t) + a(X(t) - z)|^2.$$

Differentiating and using convexity gives

$$\mathcal{E}_z' = t(3-\alpha)F - at\big(\langle X - z, \nabla F(X)\rangle - F\big) \le 0. \tag{1.6}$$

Thus $\mathcal{E}_z$ has a finite limit, $F(X) = O(t^{-2})$, and $w = tX' + a(X - z)$ is bounded. Since

$$\big(t^a(X - z)\big)' = t^{a-1}w,$$

integration from zero proves boundedness of $X$. Finite-dimensional compactness supplies cluster points, all of which minimize $F$.

For two minimizers $z_1, z_2$, set $h = |X - z_1|^2 - |X - z_2|^2$. Cancellation yields

$$\mathcal{E}_{z_1} - \mathcal{E}_{z_2} = \frac{a}{2}(th' + ah).$$

Hence $th' + ah \to \ell$ for some real $\ell$. The integrating factor $t^a$ shows $h \to \ell/a$: split the defining integral at a fixed late time and use the supremum of the tail error. If $z_1, z_2$ are cluster points, $h$ has subsequential limits $-|z_1 - z_2|^2$ and $|z_1 - z_2|^2$. They must agree. Thus the bounded trajectory has only one cluster point and converges. ∎

This direct proof includes equality $\alpha = 3$ and uses finite-dimensional compactness.

## 1.3 A separate example at low damping

Choose $\rho \in C^\infty(\mathbb{R})$ nonnegative, positive on $(0, 2)$, supported on $[0, 2]$, and flat at the endpoints. Define $g'' = \rho$ with $g(0) = g'(0) = 0$. Then $g$ is convex and smooth, $g^{-1}(0) = (-\infty, 0]$, and every positive-order derivative of $g$ is bounded. Start at $x(0) = 1$, $x'(0) = 0$.

As long as $x > 0$, the integral formula gives $x' < 0$. After any sufficiently short initial interval, the integral $\int_0^t v^\alpha g'(x(v))\,\mathrm{d}v$ has a fixed positive lower bound as long as $x$ remains positive. If $0 < \alpha \le 1$, remaining positive forever would therefore imply $x'(t) \le -Ct^{-\alpha}$ and contradict positivity. At the finite first crossing of zero, the velocity is strictly negative. Thereafter the force vanishes and

$$x'(t) = x'(T)(T/t)^\alpha,$$

so $x(t) \to -\infty$. This argument has no division by $\alpha - 1$ and includes $\alpha = 1$. Together with Proposition 1.2, the construction in Appendix A and Section 2, it proves Theorem 1.1.

### 1.4 Context and organization

The subcritical objective rate in [1] does not imply point convergence. The critical point-convergence result in [2] and the convergent infinite-length example in [3] concern distinct conclusions. The global construction in Appendix A resolves the fixed-parameter smooth cold problem addressed here; it does not supply a tame counterexample.

The positive results below concern the actual equation on an eventual valley chart, with one tangential coordinate $s$ and one normal coordinate $y$. They use confinement as an explicit hypothesis. Section 3 proves one reduced energy and virial lemma, used in all three models. Sections 4 and 5 treat nondegenerate normal oscillations; Section 6 treats exact higher-order homogeneous oscillations with a finite-time preparation hypothesis for strict action positivity. The geometry and the range of validity are part of each statement.

## 2 The full minimizer segment as omega-limit

The construction in Appendix A first imposes a central-target restriction to avoid impacts during its steering interval. A reflected energy estimate removes that restriction. The subsequent shape selection, smoothing and actual-tail insertion can then use targets approaching either endpoint of the minimizer segment.

**Lemma 2.1** (Reflected harmonic decay)**.** *Fix $\alpha > 0$ and boundaries $b_- < 0 < b_+$. Suppose that a cold reflected trajectory $z$ exists on $[0, L]$, with finitely many impact episodes on every bounded interval, and solves*

$$z'' + \frac{\alpha}{\tau} z' + (1 - d(\tau))z = 0, \qquad 0 \le d \le 1, \qquad \int_0^L d(\tau)\,\mathrm{d}\tau \le 1, \tag{2.1}$$

*between elastic impacts. At an impact the velocity changes sign, and at zero-speed contact the inward continuation is used. If $z(0) = z_0$ and $z'(0) = 0$, then*

$$|z(\tau)| + |z'(\tau)| \le C_\alpha |z_0| \tau^{-\alpha/2} \qquad (1 \le \tau \le L). \tag{2.2}$$

*The constant does not depend on the boundary positions or on $L$.*

**Proof.** Put $\mathscr{E} = (z'^2 + z^2)/2$. This is continuous at reflection and, between impacts,

$$\mathscr{E}' = -\alpha z'^2/\tau + dzz' \le d\mathscr{E}.$$

Thus $\mathscr{E} \le e\mathscr{E}(0)$ on the whole finite prefix. Let $c = \alpha/2$, $\tau_0 = \max(1, \alpha)$ and

$$\mathscr{H} = \mathscr{E} + \frac{c}{\tau} zz'.$$

For $\tau \ge \tau_0$, one has $\mathscr{E}/2 \le \mathscr{H} \le 3\mathscr{E}/2$ and

$$\mathscr{H}' + \frac{\alpha}{\tau}\mathscr{H} = dzz' + \frac{cd}{\tau} z^2 - \frac{c}{\tau^2} zz' \le C_\alpha (d + \tau^{-2})\mathscr{H}. \tag{2.3}$$

At a nonzero impact,

$$\mathscr{H}_+ - \mathscr{H}_- = -\frac{\alpha}{\tau} z z'_- \le 0,$$

because the incoming velocity has the sign of the boundary coordinate. At a grazing or resting contact there is no jump. Multiply (2.3) by

$$\tau^\alpha \exp\left(-C_\alpha \int_{\tau_0}^{\tau} (d(v) + v^{-2})\,\mathrm{d}v\right).$$

Integrating on each smooth leg and retaining the favorable jumps gives $\tau^\alpha \mathscr{H} \le C_\alpha \mathscr{E}(0)$. The preliminary energy bound handles $1 \le \tau \le \tau_0$. This proves (2.2), including the stationary case. ∎

The lemma is an a priori estimate. Existence and the finite-contact property in its use below come from Appendix A, Lemmas 4 and 6, in the range $1 < \alpha < 3$; no separate reflected existence assertion for all $\alpha > 0$ is needed.

## 2.1 Removing the central-target restriction

We use the notation of Appendix A, Section 5. Fix $1 < \alpha < 3$, $A > 0$, $s_0 \in [-S, S]$ and any $m \in (-S, S)$. Choose the local convex block centered at $m$, with narrow scale $r$, core radius $0 \le \ell < r$, tangential profile $\psi$ and fixed positive $\kappa$. Work from the beginning with its hard-wall trajectory, with kinetic masses $(A^{-2}, 1)$ and cold data $(s_0, 1)$. Let

$$z = s - m, \qquad \omega = A\sqrt{2\kappa r}, \qquad \tau = \omega t, \qquad \beta = \frac{2\alpha}{3}. \tag{2.4}$$

Between impacts its equations are

$$u'' + \alpha u'/t + V_u = 0, \qquad z'' + \alpha z'/t + \omega^2 \psi(u/r) z = 0.$$

Set

$$Q(t) = \omega \int_0^t (1 - \psi(u(v)/r))\,\mathrm{d}v, \qquad J = z^2 + z_\tau^2.$$

Both $J$ and the full mechanical energy are continuous at impacts. Thus the same differential estimate as in Appendix A gives $J \le z(0)^2 e^Q$ and, while $Q \le 1$, the tangential kinetic energy is $O(r)$.

For completeness, the estimates that survive reflection are as follows. Write

$$E = \tfrac{1}{2}u'^2 + \tfrac{1}{2}A^{-2}z'^2 + V, \qquad H = E + \frac{\beta}{t} u u'.$$

The exact identity is

$$H' + \frac{\beta}{t} H = \frac{\beta}{t}(V - u V_u - A^{-2} z'^2) - \frac{2\alpha(\alpha+3)}{9t^2} u u'. \tag{2.5}$$

The bracket is $O(r)$ on the bootstrap interval. Reflection changes neither $u$ nor $u'$, and preserves $E$, so $H$ has no jump. The initialization against the sign-force scalar oscillator, and integration of (2.5), therefore give

$$ct^{-\beta} \le E(t) \le Ct^{-\beta} \tag{2.6}$$

for $t_0 \le t \le T$ when $rT^\beta$ is small. All constants here may depend on the fixed stage but are uniform over $\ell < r$.

The occupation estimate also survives unchanged. On $|u| \le 3r$, the potential and tangential kinetic energy are $O(r)$, whence $u'^2 \ge ct^{-\beta}$. Each passage has duration at most $Crt_k^{\beta/2}$. At its outward edge, the normal speed is at least $ct_k^{-\beta/2}$; bounded outer acceleration supplies a subsequent outward interval of length at least $ct_k^{-\beta/2}$. The intervals assigned to different passages are disjoint. Their integrals of $t^\beta$ bound the passage lengths after multiplication by $Cr$. The final incomplete passage and at most one truncated assigned interval contribute at most $CrT^{\beta/2}$. The finitely many initial pieces contribute $O(r)$. Hence

$$|\{t \in [0,T] : |u(t)| \le 3r\}| \le Cr(1+T^{\beta+1}), \qquad Q(T) \le Cr^{3/2}(1+T^{\beta+1}). \tag{2.7}$$

Tangential reflection changes neither the normal speed nor the outer normal force, so it affects none of these assignments.

Choose

$$\frac{\alpha}{3} < q < \frac{3\alpha}{2\alpha+3}, \qquad T_r = r^{-q/\beta}. \tag{2.8}$$

This interval is nonempty precisely in the present subcritical range. Then

$$rT_r^\beta \to 0, \qquad Q(T_r) \to 0, \qquad \omega T_r \to \infty.$$

The same first-exit argument closes the energy and occupation bootstrap. There is no tangential strip exit: choose the strip to contain the whole wall interval. Apply Lemma 2.1 with $d = 1 - \psi(u/r)$ and boundaries $-S-m, S-m$. We obtain

$$\begin{aligned} &|s(T_r) - m| \le C(\omega T_r)^{-\alpha/2} \longrightarrow 0, \\ &T_r|s'(T_r)| \le C(\omega T_r)^{1-\alpha/2}, \qquad |u(T_r)| \le Cr^q. \end{aligned} \tag{2.9}$$

At an observation coinciding with an impact, the position is unambiguous and either velocity trace has the same magnitude. The later construction uses only the position observation. In particular, no vanishing weighted speed is asserted when $\alpha \le 2$.

## 2.2 Compatibility with completion and cold insertion

The replacement above removes only the no-impact assertion of Appendix A, Lemma 8. Its position and normal-size conclusions hold for every fixed interior target. We spell out why the later steps do not require centrality.

Fix once and for all the physical wall $W$, a guard $[-S-\eta, S+\eta]$, and a construction width $M$ larger than both $2S+\eta$ and $2e^{1/2}S$, with strict margins. Every stage strip $|s-m|<M$, $|m|\le S$, contains this guard. The convex padding and derivative bounds of Appendix A, Sections 3 and 7, use this common geometry. The wall can satisfy $W(S+\eta)>2$. Summable gradient budgets and vanishing of every block at $(0,0)$ bound the initial energy of every partial sum and of the limit by $1/4$, so the actual trajectories stay in the same strict guard even during wall penetration.

For each fixed stage the target is strictly interior. The normal zero-count argument, the strict scalar decoupling barrier and the nonzero free normal endpoint in Appendix A, Section 6, are unchanged. The smoothing, supporting-plane extension, Moreau envelope and centered convolution preserve the same exact inner product patch and complete outer half-strips. The block symmetry about $s=m$ preserves the inward tangential sign. Observation at an impact requires only position dependence, already provided by the elastic limit.

In the actual-tail insertion of Appendix A, Section 9, the data at each fixed late physical time are those of the original cold perturbed trajectory. They tend to the actual old data as the new height decreases. Near either wall a fixed interior target still supplies a strictly inward new force. Its lower bound may depend on this stage. The release estimates retain a vanishing weighted kick even when the instantaneous rescaled entry velocity is large. At a fixed small positive rescaled time, the limiting cold well state is strictly interior and the actual normalized energy is bounded. Only then is the elastic penalty limit applied. First-exit uniqueness identifies the surrogate with the actual trajectory through its certificate time; the same-sign normal certificate gives the next exact free tail.

Thus all constants that deteriorate as $m$ approaches a wall are fixed at the current stage before its physical height is lowered. No uniform positive distance of all future targets from the walls is required. This is exactly the order of choices of Appendix A, Propositions 7.1 and 9.1.

**Theorem 2.2** (Full minimizing omega-limit)**.** *For each fixed $1<\alpha<3$, the global smooth construction of Appendix A can be chosen to satisfy* (1.4) *and* (1.3)*.*

**Proof.** Use stage targets

$$m_n=(-1)^n S\left(1-\frac{1}{n+1}\right), \qquad n\ge 1,$$

and observation tolerances tending to zero. At stage $n$, choose its normalized shape, core radius, smoothing and finite certificate times first. Then lower its height to meet the insertion requirement, the derivative budgets, every earlier finite-prefix force budget, and an observation time beyond $n$ and the previous certificate. There are only finitely many requirements at each stage, and the available height has no positive lower bound.

The direct cold dependence estimate in Appendix A, Lemma 1, compares the final objective with each partial sum from the same initial data at zero. The entire future force tail is controlled by the stage's budget, so each selected observation persists in the final cold trajectory. The summable derivative bounds give one globally $C^\infty$ convex objective with all positive-order derivatives bounded. The nested zero cores and the strictly positive complete outer half-strips give its exact minimizer set $[-S, S]\times\{0\}$. The fixed wall bounds $s$, and the first block's affine normal tails bound $y$. The bounded-orbit energy argument of Appendix A, Lemma 7, gives $E(t) \to 0$, hence $y(t) \to 0$ and every cluster point lies on that segment.

The two parity subsequences of observations approach $(S, 0)$ and $(-S, 0)$. For any $\xi \in (-S, S)$, sufficiently late consecutive opposite observations lie on opposite sides of $\xi$. Continuity supplies intervening times tending to infinity with $s = \xi$. Since $y \to 0$, $(\xi, 0)$ is a cluster point. This proves equality of the omega-limit and the entire minimizer segment. ∎

# 3 A perturbed reduced energy lemma

The three valley models lead to the same one-dimensional effective dynamics. We record the needed perturbation argument explicitly; no theorem for constant damping is transferred to vanishing damping without checking its hypotheses.

**Lemma 3.1** (Energy minimization with integrable errors)**.** *Let $I$ be an open interval and $V \in C^1(I)$ be convex with a minimizer $z \in I$. Suppose $s, v$ are locally absolutely continuous on $[\tau_0, \infty)$, $s$ stays in a compact subinterval of $I$, $v$ is bounded, and*

$$\dot s = v + e, \qquad \dot v = -\delta(\tau)v - K(\tau)V'(s) + r \tag{3.1}$$

*almost everywhere. Assume $e, r \in L^1$, $K$ is locally absolutely continuous,*

$$\int_{\tau_0}^{\infty} |\dot K|\,\mathrm{d}\tau < \infty, \qquad K(\tau) \longrightarrow K_\infty > 0,$$

*and either $\delta(\tau) = \gamma/\tau$ with $\gamma > 0$, or $\delta(\tau) = \delta_0 > 0$. Then*

$$v(\tau) \longrightarrow 0, \qquad V(s(\tau)) \longrightarrow V(z). \tag{3.2}$$

*If $z$ is the unique minimizer of $V$ in $I$, then $s(\tau) \to z$.*

**Proof.** Increase $\tau_0$ so that $\tau_0 \ge 1$ and $K$ is bounded below by a positive constant. Put $U = V - V(z) \ge 0$ and

$$\mathcal{L} = \tfrac{1}{2}v^2 + KU(s).$$

The exact identity

$$\dot{\mathcal{L}} = -\delta v^2 + g, \qquad g = vr + KV'(s)e + \dot K U(s) \tag{3.3}$$

has $g \in L^1$. Indeed, all other factors are bounded on the stipulated compact set. The function $\mathcal{L} - \int_{\tau_0}^{\tau} g$ is nonincreasing and bounded below. Hence $\mathcal{L}$ has a finite nonnegative limit and $\int \delta v^2 < \infty$. In either damping case,

$$\int_{\tau_0}^{\infty} \frac{v^2}{\tau}\,\mathrm{d}\tau < \infty. \tag{3.4}$$

For $B = (s-z)v$ and $h = (s-z)^2/2$,

$$\dot{B} = v^2 + ev - \delta B - K(s-z)V'(s) + (s-z)r, \tag{3.5}$$

$$B = \dot{h} - (s-z)e. \tag{3.6}$$

Divide (3.5) by $\tau$ and integrate to an arbitrary terminal time. The kinetic integral is bounded by (3.4), and the error integrals converge absolutely. Boundedness of $B$ gives

$$\int \frac{\dot{B}}{\tau}\,\mathrm{d}\tau = \frac{B}{\tau} + \int \frac{B}{\tau^2}\,\mathrm{d}\tau$$

with uniformly bounded boundary and integral terms. For $f = \delta/\tau$, equation (3.6) gives

$$\int fB\,\mathrm{d}\tau = fh - \int f'h\,\mathrm{d}\tau - \int f(s-z)e\,\mathrm{d}\tau.$$

Here $f$ is $\gamma/\tau^2$ or $\delta_0/\tau$, so $f'$ is integrable and all displayed terms are uniformly bounded. Convexity gives $(s-z)V'(s) \geq U(s) \geq 0$. It follows that

$$\int_{\tau_0}^{\infty} \frac{KU(s)}{\tau}\,\mathrm{d}\tau < \infty, \qquad \int_{\tau_0}^{\infty} \frac{\mathcal{L}}{\tau}\,\mathrm{d}\tau < \infty.$$

The existing nonnegative limit of $\mathcal{L}$ must therefore be zero. Since $K_\infty > 0$, this proves (3.2). With a unique minimizer, compactness makes every cluster point of $s$ equal to $z$ and proves point convergence. ∎

The virial integrals above need only be uniformly bounded as signed integrals. Absolute integrability of $\dot{B}/\tau$ or $\delta B/\tau$ is neither claimed nor required. The lemma is an explicit extension of the familiar energy–virial method for convex equations with nonintegrable damping; see [4, 5] for the distinction between energy minimization and point convergence.

# 4 Exact normally quadratic valleys

Let $I$ be an open interval, let $a \in C^3(I)$ be positive, and consider the exact local objective

$$F(s,y) = a(s)y^2. \tag{4.1}$$

Its Hessian has determinant $2y^2(aa'' - 2(a')^2)$ and normal diagonal entry $2a > 0$. Consequently it is convex on $I \times \mathbb{R}$ exactly when

$$aa'' \geq 2(a')^2. \tag{4.2}$$

This is normal nondegeneracy along an entire minimizing valley, not strong convexity of $F$.

**Theorem 4.1** (Quadratic action and curvature selection)**.** *Suppose* (4.2) *holds,* $\alpha > 1$*, and a classical solution of*

$$s'' + \alpha s'/t + a'(s)y^2 = 0, \qquad y'' + \alpha y'/t + 2a(s)y = 0 \tag{4.3}$$

*on* $[T, \infty)$*,* $T > 0$*, has* $s(t)$ *in a compact subinterval* $J \Subset I$*. Then its full energy satisfies*

$$E = \tfrac{1}{2}(s'^2 + y'^2) + a(s)y^2 = O(t^{-\alpha}).$$

*If* $\alpha > 2$*, the trajectory has finite length and converges to a point of* $J \times \{0\}$*. For* $1 < \alpha \le 2$*, assume the normal solution is nontrivial and* $a$ *attains its minimum in* $I$*. Then*

$$a(s(t)) \longrightarrow \min_I a. \tag{4.4}$$

*If the minimizer of* $a$ *is unique, say* $s_*$*, then* $(s(t), y(t)) \to (s_*, 0)$*. In the nontrivial-normal case the rescaled normal action also has a strictly positive limit:*

$$t^\alpha \frac{y'^2/2 + a(s)y^2}{\sqrt{a(s)}} \longrightarrow K_\infty \in (0, \infty). \tag{4.5}$$

The confinement hypothesis is explicit. A solution of a global objective is covered only if its equation is exactly (4.3) on the entire stated tail. Neither a global completion nor entry into this valley is asserted.

**Proof.** Write $p = s'$, $u = y'$, and $A = a(s)$. All coefficient bounds below are uniform on $J$, and $A$ has a positive lower bound.

*Full energy.* Since $E' = -\alpha(p^2 + u^2)/t$ and $|yu| \le CE$, the corrected energy

$$H = E + \frac{\alpha}{2t} yu$$

is comparable to $E$ for large $t$. Direct differentiation yields the exact identity

$$H' + \frac{\alpha}{t} H = -\frac{\alpha}{2t} p^2 - \frac{\alpha}{2t^2} yu. \tag{4.6}$$

Thus $H' + \alpha H/t \le CH/t^2$. Gronwall's inequality gives $E = O(t^{-\alpha})$ and

$$|p| + |u| + |y| = O(t^{-\alpha/2}), \qquad s'' = O(t^{-1-\alpha/2} + t^{-\alpha}). \tag{4.7}$$

For $\alpha > 2$ the speed is integrable, proving the finite-length conclusion.

*Positive corrected action.* Set

$$E_N = \tfrac{1}{2}u^2 + Ay^2, \quad J_N = E_N/\sqrt{A}, \quad D = u^2 - 2Ay^2 = (yu)' + \alpha yu/t, \tag{4.8}$$

and

$$C = \frac{\alpha}{2t\sqrt{A}} + \frac{a'p}{4A^{3/2}}.$$

Exact differentiation gives

$$J_N' + \frac{\alpha}{t} J_N = -CD.$$

Therefore $\widetilde{J} = J_N + Cyu$ satisfies

$$\widetilde{J}' + \frac{\alpha}{t}\widetilde{J} = C'yu. \tag{4.9}$$

For clarity, the derivative after substitution of (4.3) is

$$C' = -\frac{\alpha}{2t^2\sqrt{A}} - \frac{\alpha a' p}{2tA^{3/2}} + \frac{a''p^2}{4A^{3/2}} - \frac{3(a')^2p^2}{8A^{5/2}} - \frac{(a')^2y^2}{4A^{3/2}}.$$

In particular

$$C = O(t^{-1} + t^{-\alpha/2}), \qquad C' = O(t^{-2} + t^{-1-\alpha/2} + t^{-\alpha}) \in L^1. \tag{4.10}$$

The normal solution cannot have $y = u = 0$ at a finite time unless it is identically zero. Also $|yu| \le J_N/\sqrt{2}$. Hence, for nontrivial normal data and sufficiently large $t$,

$$0 < \tfrac{1}{2} J_N \le \widetilde{J} \le \tfrac{3}{2} J_N.$$

For $K = t^\alpha \widetilde{J}$, equation (4.9) now gives

$$|(\log K)'| \le \sqrt{2}|C'|.$$

Its logarithm has a finite real limit. Consequently $K \to K_\infty \in (0, \infty)$ and $\int |K'| < \infty$. Finally $t^\alpha Cyu \to 0$, so (4.5) follows. This reasoning also proves the action statement when $\alpha > 2$.

*Exact tangential reduction.* For the selection assertion suppose $1 < \alpha \le 2$. Define

$$V = \sqrt{a}, \qquad b = \frac{a'}{2a}, \qquad w = p - \frac{b}{2} yu.$$

The force decomposition $a'y^2 = V' J_N - bD/2$ and (4.8) give exactly

$$w' + \frac{\alpha}{t} w + t^{-\alpha} K V'(s) = R, \qquad R = \left( V'C - \frac{b'p}{2} \right) yu. \tag{4.11}$$

Thus

$$R = O(t^{-\alpha-1} + t^{-3\alpha/2}), \qquad s' = w + \frac{b}{2} yu. \tag{4.12}$$

No oscillatory term has been discarded.

For $1 < \alpha < 2$, set

$$q = 1 - \alpha/2, \quad \tau = t^q/q, \quad v = t^{\alpha/2} w, \quad \gamma = \frac{\alpha}{2-\alpha}.$$

Since $\mathrm{d}\tau = t^{-\alpha/2}\,\mathrm{d}t$, the equations become

$$\dot{s} = v + e, \qquad \dot{v} = -\frac{\gamma}{\tau} v - K V'(s) + r, \quad e = t^{\alpha/2} \frac{b}{2} yu, \quad r = t^\alpha R. \tag{4.13}$$

Here $v$ is bounded, $K$ retains finite total variation and its positive limit, and

$$\int |e|\,\mathrm{d}\tau \le C\int_T^\infty t^{-\alpha}\,\mathrm{d}t < \infty,$$
$$\int |r|\,\mathrm{d}\tau \le C\int_T^\infty (t^{-1-\alpha/2}+t^{-\alpha})\,\mathrm{d}t < \infty.$$

At $\alpha = 2$, instead take $\tau = \log t$ and $v = tw$. The same definitions of $e, r$ give (4.13) with damping 1 in place of $\gamma/\tau$; $e, r = O(t^{-1})$ are integrable in $\tau$.

Condition (4.2) implies

$$V'' = \frac{2aa'' - (a')^2}{4a^{3/2}} \ge 0.$$

Apply Lemma 3.1 at any minimizer of $V$, equivalently of $a$. It gives (4.4). A unique minimizer and compactness give $s \to s_*$, while (4.7) gives $y \to 0$. ∎

**Example 4.2.** On $I = (-1, 1)$, $a(s) = (1 - s^2)^{-1}$ satisfies

$$aa'' - 2(a')^2 = 2(1 - s^2)^{-3} > 0.$$

Its unique curvature minimum is at zero. Any trajectory satisfying the confinement and nontrivial-normal hypotheses therefore selects $(0, 0)$ for $1 < \alpha \le 2$. This is a local example; the coefficient diverges at the endpoints of $I$.

If $y \equiv 0$, the tangential motion has $s' = ct^{-\alpha}$ and may converge to any point of the valley. The normal nontriviality condition for selection is thus essential. If $a$ has a nontrivial interval of minimizers, the theorem asserts convergence of the coefficient value, not point convergence or finite length in the selection range.

# 5 Analytic normally nondegenerate valleys

The exact quadratic normal profile can be replaced by analytic higher-order terms. The relevant action is then the action of the full frozen nonlinear oscillator. Its first-order oscillatory drift can still be removed by explicit correctors. Action correctors and near-identity averaging are classical tools [6, 7]; action-weighted effective frequency potentials also appear in constrained and high-frequency Hamiltonian dynamics [8, 9]. We give the estimates needed for the present damped equation directly.

**Theorem 5.1** (Confined analytic valleys)**.** *Let $I \subset \mathbb{R}$ be an open interval, let $\rho > 0$, and let $F$ be real analytic and convex on*

$$D = I \times (-\rho, \rho).$$

*Assume, for every $s \in I$, that*

$$F(s, 0) = F_y(s, 0) = 0, \qquad a(s) = \tfrac{1}{2}F_{yy}(s, 0) > 0. \tag{5.1}$$

*Let $\alpha > 1$, and let $(s, y)$ be a solution of the actual gradient equation*

$$s'' + \frac{\alpha}{t}s' + F_s(s, y) = 0, \qquad y'' + \frac{\alpha}{t}y' + F_y(s, y) = 0. \tag{5.2}$$

*Suppose explicitly that there exist $T_0 > 0$, a compact subinterval $J \Subset I$, and $0 < \rho_0 < \rho$ such that*

$$(s(t), y(t)) \in J \times [-\rho_0, \rho_0] \qquad (t \geq T_0). \tag{5.3}$$

*Then the following assertions hold.*

1. *The mechanical energy satisfies*

$$E(t) = \tfrac{1}{2}\big(s'(t)^2 + y'(t)^2\big) + F(s(t), y(t)) = O(t^{-\alpha}).$$

2. *The exact frozen normal action $\mathcal{I}(t)$, defined for all sufficiently large $t$ by (5.8), satisfies*

$$t^\alpha \mathcal{I}(t) \longrightarrow K_\infty \in [0, \infty).$$

   *There is a corrected rescaled action $K(t)$ with finite total variation and the same limit. If $(y, y')$ is not identically $(0, 0)$, then $K_\infty > 0$.*

3. *If $\alpha > 2$, the trajectory has finite tail length and converges to a point of $J \times \{0\}$.*

4. *If $1 < \alpha \leq 2$, the normal motion is nontrivial, and $a$ attains its minimum on $I$, then*

$$a(s(t)) \longrightarrow \min_I a.$$

   *If $a$ is nonconstant, its minimizer $m$ is unique and $(s(t), y(t)) \to (m, 0)$.*

5. *If $a$ is constant on $I$, then every trajectory satisfying (5.3) converges to a point of $J \times \{0\}$, and*

$$s'(t) = O(t^{-\alpha}).$$

*If the normal motion is identically zero, then $s'(t) = ct^{-\alpha}$ and the trajectory converges, without a curvature-selection conclusion. In particular, the existence of a minimizer of $a$, together with (5.3), implies point convergence for every $\alpha > 1$.*

All conclusions concern the actual solution on the stated positive-time tail. Confinement is a hypothesis; no entry into the strip or preservation of the confinement rectangle is asserted. No smallness of the initial energy is required under this hypothesis.

## 5.1 Energy decay before oscillator reduction

Write $p = s'$, $u = y'$ and $d(t) = \alpha/t$. Constants below are uniform on the compact sets under consideration; the starting time may be increased finitely many times.

Convexity and (5.1) imply $F \geq 0$ on $D$. For each fixed $s$, positive normal curvature at zero and convexity give $F(s, y) > 0$ for $y \neq 0$. The function $F(s, y)/y^2$ extends continuously to $y = 0$, with value $a(s) > 0$. Compactness therefore gives

$$F(s, y) \geq cy^2 \quad \text{on } J \times [-\rho_0, \rho_0] \tag{5.4}$$

for some $c > 0$.

Energy dissipation gives

$$E' = -\frac{\alpha}{t}(p^2 + u^2), \qquad \int_{T_0}^{\infty} \frac{p^2 + u^2}{t}\,\mathrm{d}t < \infty.$$

In particular, $p, u$ are bounded. Fix $s_0 \in I$, put $X_0 = (s_0, 0)$ and let $b = |(s, y) - X_0|^2/2$. Both $b$ and $b'$ are bounded, and convexity yields

$$b'' + \frac{\alpha}{t}b' = p^2 + u^2 - \langle (s, y) - X_0, \nabla F(s, y)\rangle \leq p^2 + u^2 - F(s, y).$$

After division by $t$ and integration over $[T_0, R]$, the terms involving $b$ are bounded uniformly in $R$: indeed,

$$\int_{T_0}^{R} \frac{b''}{t}\,\mathrm{d}t = \left[\frac{b'}{t}\right]_{T_0}^{R} + \int_{T_0}^{R} \frac{b'}{t^2}\,\mathrm{d}t.$$

It follows that $\int_{T_0}^{\infty} F(s, y)\,\mathrm{d}t/t < \infty$, hence that $\int_{T_0}^{\infty} E(t)\,\mathrm{d}t/t < \infty$. Since $E$ is nonnegative and decreasing, $E(t) \to 0$. Thus $y, p, u \to 0$, and the trajectory eventually enters a uniformly small normal neighborhood.

To obtain the full rate, define

$$H = E + \frac{d}{2}yu.$$

Direct differentiation gives the exact identity

$$H' + dH = -\frac{d}{2}p^2 + \frac{d}{2}(2F - yF_y) + \frac{d'}{2}yu. \tag{5.5}$$

By (5.4), $|yu| \leq CE$, so $H/E = 1 + O(t^{-1})$ whenever $E > 0$. Analyticity gives $2F - yF_y = O(|y|^3)$ uniformly on $J$. Consequently, on a sufficiently late tail,

$$H' + \frac{\alpha}{t}H \leq \frac{C}{t}H^{3/2} + \frac{C}{t^2}H. \tag{5.6}$$

If $E$ vanishes at a finite time, the solution is stationary and the conclusions follow directly. Otherwise $H > 0$ eventually. Since $H \to 0$, for any $0 < \beta < \alpha$ the logarithmic form of (5.6) first gives

$$(\log H)' \leq -\frac{\beta}{t}$$

eventually, hence $H = O(t^{-\beta})$. Fix one such $\beta$. Then $\int^\infty \sqrt{H}\,\mathrm{d}t/t < \infty$, so a second integration of (5.6), now retaining the full coefficient $\alpha$, gives $H = O(t^{-\alpha})$. We have proved

$$\begin{aligned} E &= O(t^{-\alpha}), \\ |p| + |u| + |y| &= O(t^{-\alpha/2}), \\ |p'| &\le C\big(t^{-1-\alpha/2} + t^{-\alpha}\big). \end{aligned} \tag{5.7}$$

The last estimate follows from $F_s = O(y^2)$ and the tangential equation. For $\alpha > 2$, the velocity bound is integrable, proving finite tail length and convergence to a valley point.

## 5.2 The exact normal action and its homological inverse

Choose a compact interval $J_1 \Subset I$ whose interior contains $J$. For frozen $s \in J_1$, define the normal energy $N(s, y, u) = u^2/2 + F(s, y)$. Sufficiently small positive energy levels are periodic ovals around $(y, u) = (0, 0)$. Their action and the phase-space action are

$$\begin{gathered} A(s,\eta) = \frac{1}{2\pi} \oint_{N=\eta} u\,\mathrm{d}y, \qquad \mathcal{I}(s,y,u) = A(s, N(s,y,u)), \\ \omega(s) = \sqrt{2a(s)}, \end{gathered} \tag{5.8}$$

where the integral follows the Hamiltonian orientation.

Here is a uniform analytic construction near the equilibrium. The map

$$z = y\sqrt{\frac{2F(s,y)}{y^2}}$$

uses the analytic extension of $F/y^2$ at zero and the positive square root. It has $z_y(s, 0) = \omega(s) > 0$, and therefore has an analytic inverse $y = Y(s, z)$ on a common neighborhood of $J_1 \times \{0\}$. Put

$$z = r\cos\theta, \qquad u = r\sin\theta, \qquad r = \sqrt{2\eta}.$$

The action and the period $P$ are

$$\begin{aligned} A(s,\eta) &= \frac{r^2}{2\pi} \int_0^{2\pi} \sin^2\theta\, Y_z(s, r\cos\theta)\,\mathrm{d}\theta, \\ P(s,\eta) &= \int_0^{2\pi} Y_z(s, r\cos\theta)\,\mathrm{d}\theta. \end{aligned} \tag{5.9}$$

Odd powers of $r$ vanish on integration. Thus $A$ and $P$ are analytic in $(s, \eta)$ near $\eta = 0$. Differentiation of the action integral with respect to energy gives $2\pi A_\eta = P$, and the leading terms give

$$A(s,\eta) = \frac{\eta}{\omega(s)} + O(\eta^2), \qquad P = 2\pi A_\eta. \tag{5.10}$$

The local inverse $\eta = h(s, \mathcal{I})$ is analytic and satisfies

$$h(s,\mathcal{I}) = \omega(s)\mathcal{I} + O(\mathcal{I}^2), \qquad h_{\mathcal{I}} = 1/A_\eta. \tag{5.11}$$

These estimates hold uniformly with the required $s$ derivatives. In particular, $\mathcal{I}$ is comparable to $r^2$, and the frozen frequency is bounded away from zero.

Let

$$L_s = u\partial_y - F_y(s,y)\partial_u$$

be the frozen normal generator. The polar equations are $z' = u/Y_z$, $u' = -z/Y_z$, and hence

$$\theta' = -\frac{1}{Y_z(s, r\cos\theta)}. \tag{5.12}$$

This formula gives the following uniform inverse on zero-mean functions.

**Lemma 5.2** (Polar homological estimate)**.** *Let $j(s,y,u)$ be analytic near $J_1 \times \{(0,0)\}$ and vanish to normal order at least $\ell$, where $\ell \geq 2$ is an integer. Assume that its frozen time average is zero on every sufficiently small positive-energy oval. Then $L_s\chi_j = j$ has a single-valued periodic solution satisfying*

$$|\chi_j| + |\partial_s\chi_j| \leq C\mathcal{I}^{\ell/2}, \qquad |\nabla_{y,u}\chi_j| \leq C\mathcal{I}^{(\ell-1)/2}. \tag{5.13}$$

*The derivative $\partial_s$ is taken at fixed physical coordinates $(y,u)$. The solution extends by zero to a $C^1$ function at the normal origin.*

**Proof.** Denote the polar pullback of $j$ by $\widehat{j}(s,r,\theta)$. Set

$$\widehat{\chi}_j(s,r,\theta) = -\int_0^\theta Y_z(s, r\cos\phi)\widehat{j}(s,r,\phi)\,\mathrm{d}\phi. \tag{5.14}$$

The zero-time-mean assumption says exactly that the full-period integral of the integrand vanishes. Thus (5.14) is periodic; its parameter and radial derivatives match at the angular seam because the full-period integral vanishes identically in $(s,r)$. Equation (5.12) proves $L_s\chi_j = j$.

Analytic normal vanishing gives

$$\widehat{\chi}_j,\ \partial_s\widehat{\chi}_j,\ \partial_\theta\widehat{\chi}_j = O(r^\ell), \qquad \partial_r\widehat{\chi}_j = O(r^{\ell-1}).$$

At fixed physical coordinates one has $z_s = O(r)$, $r_s = O(r)$ and $\theta_s = O(1)$; also $\nabla_{y,u}r = O(1)$ and $\nabla_{y,u}\theta = O(r^{-1})$. The chain rule and $\mathcal{I} \asymp r^2$ give (5.13). These bounds also give the stated $C^1$ extension, although no phase angle extends through the origin. ∎

Write $\langle\cdot\rangle$ for frozen time averaging. The two action identities are

$$\langle u^2\rangle = \frac{A}{A_\eta}, \qquad \langle F_s\rangle = -\frac{A_s}{A_\eta} = h_s(s,\mathcal{I}). \tag{5.15}$$

Indeed, $\oint u^2\,\mathrm{d}t = \oint u\,\mathrm{d}y = 2\pi A$, while differentiating the action at fixed energy gives $2\pi A_s = -\oint F_s\,\mathrm{d}t$. The latter identity can also be obtained by differentiating the one-dimensional integral between the two turning points; its endpoint terms vanish. Divide by $P = 2\pi A_\eta$ and differentiate $A(s, h(s,\mathcal{I})) = \mathcal{I}$ to obtain (5.15).

We apply Lemma 5.2 to

$$\begin{aligned} f &= A_\eta(s,N)u^2 - \mathcal{I}, \\ g &= A_s(s,N) + A_\eta(s,N)F_s(s,y), \\ k &= F_s(s,y) - h_s(s,\mathcal{I}). \end{aligned} \tag{5.16}$$

All three functions are analytic, vanish to normal order at least two, and have zero frozen mean by (5.15). Their primitives $\chi_f, \chi_g, \chi_k$ consequently satisfy

$$|\chi_j| + |\chi_{j,s}| \le C\mathcal{I}, \qquad |\nabla_{y,u}\chi_j| \le C\sqrt{\mathcal{I}} \quad (j = f, g, k). \tag{5.17}$$

The energy argument has already shown that these constructions apply along the actual trajectory for all sufficiently large times.

## 5.3 Strictly positive limiting action and bounded variation

Along the actual solution, $N' = -du^2 + pF_s$, so

$$\mathcal{I}' = -d\mathcal{I} - df + pg. \tag{5.18}$$

For any of the correctors, its actual derivative is

$$\chi' = L_s\chi + p\chi_s - du\chi_u.$$

Define

$$Q = \mathcal{I} + d\chi_f - p\chi_g, \qquad K = t^\alpha Q. \tag{5.19}$$

Substitution in (5.18) gives the exact cancellation

$$\begin{aligned} Q' + dQ = Z = {} & d'\chi_f + dp\chi_{f,s} - d^2u\chi_{f,u} \\ & - p'\chi_g - p^2\chi_{g,s} + pdu\chi_{g,u} \\ & + d^2\chi_f - dp\chi_g. \end{aligned} \tag{5.20}$$

Using (5.17) and $|u| \le C\sqrt{\mathcal{I}}$, we obtain

$$\begin{aligned} |Z| &\le C\left(t^{-2} + \frac{|p|}{t} + p^2 + |p'|\right)\mathcal{I}, \\ \frac{Q}{\mathcal{I}} &= 1 + O(t^{-1} + |p|), \end{aligned} \tag{5.21}$$

whenever $\mathcal{I} > 0$.

If the normal motion is nontrivial, it cannot reach $y = u = 0$ at a finite positive time. That set is invariant, and uniqueness of (5.2) would otherwise force the normal motion to vanish identically. Thus $\mathcal{I} > 0$ throughout the late tail, and $Q > 0$ eventually. By (5.7) and (5.21),

$$|(\log K)'| \le C\left(t^{-2} + t^{-1-\alpha/2} + t^{-\alpha}\right). \tag{5.22}$$

The right-hand side is integrable for $\alpha > 1$. Hence $\log K$ has a finite real limit, and $K \to K_\infty \in (0,\infty)$. Moreover, $K$ is bounded and (5.22) gives $\int^\infty |K'|\,\mathrm{d}t < \infty$. Thus $K$ has finite total variation. Since $Q/\mathcal{I} \to 1$, one also has $t^\alpha \mathcal{I} \to K_\infty$.

For identically zero normal motion, all three correctors vanish and $\mathcal{I} = Q = K = 0$. This proves the action assertions in every case. The use of the full nonlinear action is important: (5.22) contains no unaveraged relative error of size $O(\sqrt{\mathcal{I}})$, which need not be integrable when $\alpha \le 2$.

## 5.4 Exact tangential reduction and curvature selection

Set $w = p + \chi_k$. The identity $L_s \chi_k = F_s - h_s$ gives

$$w' + dw + h_s(s,\mathcal{I}) = R_0, \qquad R_0 = p\chi_{k,s} - du\chi_{k,u} + d\chi_k. \tag{5.23}$$

Thus $|R_0| \le C(|p| + t^{-1})\mathcal{I}$. Define

$$R = R_0 - \big(h_s(s,\mathcal{I}) - \omega'(s)\mathcal{I}\big) + \omega'(s)\big(t^{-\alpha}K - \mathcal{I}\big).$$

Equations (5.11), (5.19) and (5.17) yield the exact system

$$w' + \frac{\alpha}{t} w + t^{-\alpha} K \omega'(s) = R, \qquad s' = w - \chi_k, \tag{5.24}$$

with the complete remainder bound

$$\begin{aligned} |R| &\le C\big[(t^{-1} + |p|)\mathcal{I} + \mathcal{I}^2\big] \\ &\le C\big(t^{-\alpha-1} + t^{-3\alpha/2} + t^{-2\alpha}\big). \end{aligned} \tag{5.25}$$

In particular, no limiting averaged equation has been substituted for the actual dynamics.

For $1 < \alpha < 2$, put $q = 1 - \alpha/2$, $\tau = t^q/q$ and $v = t^{\alpha/2} w$. For $\alpha = 2$, put $\tau = \log t$ and $v = tw$. In either case $\mathrm{d}\tau = t^{-\alpha/2}\,\mathrm{d}t$. With a dot denoting $\tau$ differentiation, (5.24) becomes

$$\dot{s} = v + e, \qquad \dot{v} = -\delta(\tau) v - K\omega'(s) + r, \tag{5.26}$$

where

$$e = -t^{\alpha/2}\chi_k, \qquad r = t^\alpha R, \qquad \delta(\tau) = \begin{cases} \dfrac{\alpha}{(2-\alpha)\tau}, & 1 < \alpha < 2, \\ 1, & \alpha = 2. \end{cases} \tag{5.27}$$

The variables $s, v$ are bounded. The errors tend to zero, and their exact weighted estimates are

$$\begin{aligned} &\int^\infty |e|\,\mathrm{d}\tau \le C \int^\infty t^{-\alpha}\,\mathrm{d}t < \infty, \\ &\int^\infty |r|\,\mathrm{d}\tau \le C \int^\infty \big(t^{-1-\alpha/2} + t^{-\alpha} + t^{-3\alpha/2}\big)\,\mathrm{d}t < \infty. \end{aligned} \tag{5.28}$$

These integrability statements include $\alpha = 2$. The increasing time change preserves the finite total variation of $K$ and its positive limit when the normal motion is nontrivial.

It remains to identify the reduced potential. Expanding the convex Hessian determinant at $y = 0$ gives

$$\det D^2F(s,y) = 2\big(aa'' - 2(a')^2\big)y^2 + O(y^3) \geq 0,$$

and therefore $aa'' \geq 2(a')^2$. In particular,

$$\omega'' = \frac{2aa'' - (a')^2}{(2a)^{3/2}} \geq 0. \tag{5.29}$$

Suppose $m \in I$ minimizes $a$, and hence also $\omega$. Apply Lemma 3.1 to (5.26) with the convex nonnegative potential $V(s) = \omega(s) - \omega(m)$. Its hypotheses have all been verified: bounded $s, v$, the $L^1$ errors (5.28), positive limiting $K$ of finite total variation, and damping of the required form. It follows that

$$\frac{1}{2}v^2 + K\big(\omega(s) - \omega(m)\big) \longrightarrow 0.$$

Since $K_\infty > 0$, $\omega(s(t)) \to \omega(m)$, proving curvature selection. If $a$ is nonconstant, two distinct minima would force the convex function $\omega$ to be constant on the interval between them. Analyticity would then make $a$ constant on all of $I$. Thus the minimizer is unique. Compact confinement and $y \to 0$ now imply $(s, y) \to (m, 0)$; in particular, the hypotheses force $m \in J$.

## 5.5 The constant-curvature case

When $a$ is constant, the absence of the quadratic tangential force improves the order of the tangential corrector. Namely,

$$F_s = O(r^3), \qquad h_s = O(\mathcal{I}^2), \qquad k = O(r^3).$$

Lemma 5.2 with $\ell = 3$ therefore gives

$$|\chi_k| + |\chi_{k,s}| \leq C\mathcal{I}^{3/2}, \qquad |\nabla_{y,u}\chi_k| \leq C\mathcal{I}. \tag{5.30}$$

Using this more precise corrector in (5.23), we obtain

$$|w' + dw| \leq C\big[(|p| + d)\mathcal{I}^{3/2} + \mathcal{I}^2\big], \qquad p = w + O(\mathcal{I}^{3/2}).$$

Since $\mathcal{I} = O(t^{-\alpha})$, the quantity $z = t^\alpha w$ satisfies, for $t \geq T \geq 1$,

$$|z'| \leq Ct^{-3\alpha/2}|z| + C\big(t^{-1-\alpha/2} + t^{-\alpha}\big). \tag{5.31}$$

Both the coefficient and the forcing are integrable for $\alpha > 1$. The integral form of Gronwall's inequality gives boundedness of $z$, whence $p = O(t^{-\alpha})$. Therefore $s'$ is integrable and $s$ converges; the already proved $y \to 0$ gives convergence to a valley point. For zero normal motion, the same conclusion follows directly from $s'' + \alpha s'/t = 0$. Together with the preceding subsections, this completes the proof of Theorem 5.1.

### 5.6 A nonquadratic example and the local domain restriction

For $c \in \mathbb{R}$ and $d_0 \geq 0$, consider

$$F(s,y) = \frac{y^2}{1-s^2} + cy^3 + d_0 y^4, \qquad -1 < s < 1. \tag{5.32}$$

Direct differentiation gives

$$\begin{aligned} F_{ss} &= \frac{2(1+3s^2)y^2}{(1-s^2)^3}, \\ F_{yy} &= \frac{2}{1-s^2} + 6cy + 12d_0 y^2, \\ \det D^2 F &= \frac{4y^2}{(1-s^2)^3}\big[1 + 3c(1+3s^2)y + 6d_0(1+3s^2)y^2\big]. \end{aligned}$$

If $c \neq 0$, any $0 < \rho < 1/(12|c|)$ makes the bracket strictly positive on $(-1,1) \times (-\rho,\rho)$, while $F_{yy} \geq 2 - 6|c|\rho > 0$. Thus $F$ is convex on the full strip, with $a(s) = (1-s^2)^{-1}$ and unique curvature minimum at $s = 0$. If $c = 0$, convexity holds on $(-1,1) \times \mathbb{R}$. The theorem applies to every nontrivial-normal trajectory explicitly confined to a compact subrectangle; the example does not establish such confinement for arbitrary initial data.

The local analytic hypothesis cannot in general be replaced by global analyticity on $\mathbb{R}^2$ while retaining variable curvature along a straight minimizer segment. Indeed, let a finite globally convex real-analytic function have such a segment, and subtract its minimum. Its restriction to the containing line vanishes identically by analyticity. After a change of coordinates this is the line $y = 0$. For $0 < \lambda < 1$, convexity and

$$(s+h, y) = (1-\lambda)\left(s, \frac{y}{1-\lambda}\right) + \lambda\left(s + \frac{h}{\lambda}, 0\right)$$

give

$$F(s+h, y) \leq (1-\lambda) F\left(s, \frac{y}{1-\lambda}\right).$$

Letting $\lambda \downarrow 0$ gives $F(s+h,y) \leq F(s,y)$; reversing the displacement gives equality. Hence $F$ is invariant in the tangential direction and its normal curvature is constant. A variable-curvature model such as (5.32) is therefore genuinely local in its analytic domain. The theorem supplies neither a globally analytic variable-curvature completion nor a general convergence result for arbitrary tame convex objectives. Multiple normal modes, vanishing normal frequency, and the boundary $\alpha = 1$ require additional arguments.

## 6 Exact normally degenerate homogeneous valleys

We now consider an integer $m \geq 2$, a positive function $a \in C^2(I)$ on an open interval, and

$$\begin{aligned} s'' + \alpha s'/t + a'(s) y^{2m} &= 0, \\ y'' + \alpha y'/t + 2m a(s) y^{2m-1} &= 0. \end{aligned} \tag{6.1}$$

Unless confinement is proved explicitly below, the trajectory is assumed to satisfy $s(t) \in J \Subset I$ for $t \geq T > 0$. This is an exact local homogeneous model. It does not include a Taylor remainder or assert a globally smooth convex completion.

## 6.1 Geometry and unconditional energy decay

For $F(s, y) = a(s)y^{2m}$,

$$\det \nabla^2 F = 2my^{4m-2}\big((2m-1)aa'' - 2m(a')^2\big).$$

The normal diagonal is positive off $y = 0$, and the whole Hessian is zero on $y = 0$. Thus convexity is equivalent to

$$(2m-1)aa'' \geq 2m(a')^2, \tag{6.2}$$

or, equivalently, concavity of $a^{-1/(2m-1)}$. Under this condition the effective potential

$$V = a^{1/(m+1)} \tag{6.3}$$

is convex, since

$$V'' = \frac{a^{1/(m+1)-2}}{m+1}\left(aa'' - \frac{m}{m+1}(a')^2\right) \geq 0.$$

The example

$$I = (-1, 1), \qquad a(s) = (1 - s^2)^{-(2m-1)} \tag{6.4}$$

has its unique minimum at zero and satisfies

$$(2m-1)aa'' - 2m(a')^2 = 2(2m-1)^2(1-s^2)^{1-4m} > 0.$$

For the remainder of this section write

$$\begin{gathered} p = s', \quad u = y', \quad A = a(s), \quad E = \tfrac{1}{2}(p^2 + u^2) + Ay^{2m}, \quad E_N = \tfrac{1}{2}u^2 + Ay^{2m}, \\ b = \frac{\alpha}{m+1}, \quad \beta = 2mb, \quad q = 1 - (m-1)b, \quad d = \frac{bq}{2}, \quad \nu = \frac{2m}{m-1}, \\ H = E + \frac{b}{t}yu + \frac{d}{t^2}y^2. \end{gathered} \tag{6.5}$$

**Proposition 6.1** (Energy decay without action preparation)**.** *Suppose* $0 < \alpha < (m+1)/(m-1)$ *and a solution of* (6.1) *has* $s(t) \in J \Subset I$ *for* $t \geq T$*. Then*

$$E(t) = O(t^{-\beta}), \quad |y(t)| = O(t^{-b}), \quad |p(t)| + |u(t)| = O(t^{-mb}). \tag{6.6}$$

*Neither convexity nor a positive-action assumption is needed for this estimate. If* $\beta > 2$*, the trajectory has finite length and converges to a point of* $J \times \{0\}$*. The constants can be chosen uniformly for* $T \geq 1$ *and bounded scaled initial energy* $T^\beta E(T)$*, with* $m, \alpha$ *and the coefficient bounds on* $J$ *fixed.*

**Proof.** The full energy satisfies $E' = -\alpha(p^2 + u^2)/t$. Direct calculation gives

$$H' + \frac{\beta}{t}H = -\frac{b}{t}p^2 + \frac{d(\beta-2)}{t^3}y^2 \tag{6.7}$$

and the square completion

$$H = \tfrac{1}{2}p^2 + \tfrac{1}{2}(u + by/t)^2 + Ay^{2m} + \frac{b(1-mb)}{2t^2}y^2. \tag{6.8}$$

If $\beta \le 2$, all terms in (6.8) are nonnegative and $t^\beta H$ is nonincreasing. Consequently $y = O(t^{-b})$, $p = O(t^{-mb})$, and

$$|u| \le \sqrt{2H} + b|y|/t = O(t^{-mb}),$$

because $b + 1 > mb$, equivalently $q > 0$. This includes $\beta = 2$.

For the full stated range, $H$ need not be nonnegative. Young's inequality, using the positive lower bound of $A$, gives

$$H \ge \tfrac{1}{2}E - C_0 t^{-\nu}.$$

Indeed, first bound $b|yu|/t$ by $u^2/4 + b^2y^2/t^2$, then absorb a fixed multiple of $y^2/t^2$ into $Ay^{2m}/2 + C_0t^{-\nu}$. Increase $C_0$ if necessary and set $L = H + C_0t^{-\nu} \ge E/2 \ge 0$. Since $\beta < \nu$, equation (6.7) yields

$$L' + \frac{\beta}{t}L \le Ct^{-3}L^{1/m}.$$

The added term contributes $(\beta - \nu)C_0t^{-\nu-1} \le 0$, and $y^2 \le CL^{1/m}$. With $Z = t^\beta L$,

$$Z' \le Ct^{-1-2q}Z^{1/m}.$$

A positive regularization allows integration also at zeros and gives

$$Z(t)^{(m-1)/m} \le Z(T)^{(m-1)/m} + CT^{-2q}. \tag{6.9}$$

Thus $E \le 2L = O(t^{-\beta})$. All initial terms in $T^\beta L(T)$ are uniformly bounded for bounded $T^\beta E(T)$ and $T \ge 1$, because $q > 0$. This proves the uniformity assertion. If $\beta > 2$, integration of the speed bound gives finite length and the claimed limit. ∎

No estimate uniform as $q \downarrow 0$, and no conclusion at $\alpha = (m+1)/(m-1)$, is asserted.

## 6.2 Finite-time preparation and positive limiting action

Let

$$\theta = \frac{m+1}{2m}, \qquad \eta = \frac{m-1}{2m}, \qquad J_N = E_N^\theta A^{-1/(2m)}. \tag{6.10}$$

Up to a fixed $m$-dependent normalization this is the frozen homogeneous oscillator's action.

**Theorem 6.2** (Prepared homogeneous selection)**.** *Fix $m \geq 2$, $1 < \alpha \leq (m+1)/m$, a positive $a \in C^2(I)$, $J \Subset I$, and constants $M, k_0 > 0$. There is a time $T_0 \geq 1$, depending only on these data and the coefficient bounds on $J$, with the following property. If $T \geq T_0$, a solution of* (6.1) *stays in $J$ for $t \geq T$, and*

$$T^\beta H(T) \leq M, \qquad T^\alpha J_N(T) \geq k_0, \tag{6.11}$$

*then a correction $K$ of $t^\alpha J_N$ has finite total variation and*

$$t^\alpha J_N(t) \longrightarrow K_\infty \in (0,\infty), \qquad E_N(t) \asymp t^{-\beta}. \tag{6.12}$$

*If also* (6.2) *holds and $a$ has a minimizer in $I$, then*

$$a(s(t)) \longrightarrow \min_I a, \tag{6.13}$$

*and the more precise energy asymptotics are*

$$\begin{aligned} &t^\beta E_N(t) \longrightarrow K_\infty^{1/\theta} \min_I V, \\ &t^\beta p(t)^2 \longrightarrow 0, \qquad t^\beta E(t) \longrightarrow K_\infty^{1/\theta} \min_I V > 0. \end{aligned} \tag{6.14}$$

*A unique coefficient minimizer $s_*$ implies $(s(t), y(t)) \to (s_*, 0)$.*

The hypotheses in (6.11) concern one finite time. They do not assume the desired asymptotic action limit. The confinement assumption remains separate. For instance, bounded scaled data

$$y(T) = T^{-b} Y, \quad u(T) = T^{-mb} U, \quad p(T) = T^{-mb} P$$

with $(Y, U)$ bounded away from $(0,0)$ satisfy inequalities of this form for suitable $M, k_0$; their subsequent confinement must still be assumed or proved.

**Proof.** *Exact logarithmic correction.* Nonzero normal data give $E_N > 0$ at every finite time by uniqueness, so the following quantities are well defined:

$$D = u^2 - 2mAy^{2m} = (yu)' + \alpha yu/t, \quad c = \frac{2\alpha/t + A'/A}{4m}, \quad Q = \frac{yu}{E_N}. \tag{6.15}$$

Differentiating (6.10) gives

$$(\log J_N)' + \alpha/t = -cD/E_N.$$

It follows, without averaging, that

$$\ell = \log(t^\alpha J_N) + cQ, \qquad \ell' = \left[c' - c\left(\frac{E_N'}{E_N} + \frac{\alpha}{t}\right)\right] Q. \tag{6.16}$$

The compact coefficient bounds imply

$$|Q| \leq C E_N^{-\eta}, \qquad |E_N'/E_N| \leq C(t^{-1} + |p|), \tag{6.17}$$

$$|c'| \le C(t^{-2} + |p|/t + p^2 + E_N). \tag{6.18}$$

For the last estimate one may use the exact derivative

$$4mc' = -2\alpha/t^2 + \left(\frac{a''}{a} - \frac{(a')^2}{a^2}\right) p^2 + \frac{a'}{a} p', \qquad p' = -\alpha p/t - a' y^{2m}.$$

Thus

$$|\ell'| \le C(t^{-2} + |p|/t + p^2 + E_N) E_N^{-\eta}. \tag{6.19}$$

*A noncircular positivity bootstrap.* Since $\beta \le 2$, equations (6.7)–(6.8) and (6.11) give $E \le C(M) t^{-\beta}$, uniformly in the late starting time. On an interval where $t^\alpha J_N \ge k_0/2$, equation (6.10) also yields $E_N \ge c_0 t^{-\beta}$. Substitution in (6.17)–(6.19) gives

$$|cQ| \le C(t^{-q} + t^{-b}), \qquad |\ell'| \le C(t^{-1-q} + t^{-1-b} + t^{-\alpha}). \tag{6.20}$$

All constants are uniform in $T$ on this stopped interval. Including both endpoint corrections $cQ$ in the integral of (6.16), we obtain

$$\left| \log \frac{t^\alpha J_N(t)}{T^\alpha J_N(T)} \right| \le C(T^{-q} + T^{-b} + T^{1-\alpha}). \tag{6.21}$$

Choose $T_0$ so that the right side is less than $\log(3/2)$. A first exit from $t^\alpha J_N \ge k_0/2$ would contradict (6.21), since the initial value is at least $k_0$. There is no such exit. Hence (6.20) holds on the whole tail, $\ell$ has an absolutely convergent derivative integral and a finite real limit, and

$$K = \exp \ell, \qquad 0 < K_\infty < \infty, \qquad \int_T^\infty |K'| \, \mathrm{d}t < \infty. \tag{6.22}$$

Since $cQ \to 0$, this proves (6.12).

*Exact reduced dynamics.* Define

$$h(s) = \frac{a'(s)}{2(m+1)a(s)}, \qquad w = p - h(s) y u.$$

The force decomposition and the corrected tangential equation are

$$\begin{aligned} a' y^{2m} &= V' J_N^{1/\theta} - hD, \\ w' + \alpha w/t + V' J_N^{1/\theta} &= -h' p \, y u. \end{aligned} \tag{6.23}$$

Here $h'$ denotes differentiation with respect to $s$. Set $B = K^{1/\theta}$. Then $B$ has finite total variation and a positive limit, and

$$\begin{aligned} w' + \alpha w/t + t^{-\beta} B V'(s) &= R, \\ R &= V' t^{-\beta} B (1 - e^{-cQ/\theta}) - h' p \, y u. \end{aligned} \tag{6.24}$$

The estimates already proved imply

$$yu = O(t^{-\alpha}), \qquad R = O(t^{-\alpha-1} + t^{-(2m+1)b}). \tag{6.25}$$

If $mb < 1$, put

$$\sigma = \frac{t^{1-mb}}{1-mb}, \qquad v = t^{mb}w, \qquad \gamma = \frac{\alpha - mb}{1-mb} = \frac{b}{1-mb} > 0.$$

Since $\mathrm{d}\sigma = t^{-mb}\,\mathrm{d}t$, the exact equations are

$$\frac{\mathrm{d}s}{\mathrm{d}\sigma} = v + e, \qquad \frac{\mathrm{d}v}{\mathrm{d}\sigma} = -\frac{\gamma}{\sigma}v - BV'(s) + \rho, \quad e = t^{mb}h\,yu, \quad \rho = t^{\beta}R. \tag{6.26}$$

The variables $s, v$ are bounded, and

$$\int |e|\,\mathrm{d}\sigma \le C\int_T^\infty t^{-\alpha}\,\mathrm{d}t < \infty,$$
$$\int |\rho|\,\mathrm{d}\sigma \le C\int_T^\infty (t^{-1-b} + t^{-\alpha})\,\mathrm{d}t < \infty.$$

At $mb = 1$, use $\sigma = \log t$ and $v = tw$. The damping in (6.26) is then the positive constant

$$\alpha - 1 = b = 1/m.$$

The same integral estimates hold. In both cases apply Lemma 3.1 to $V$ and $B$. It gives $V(s) \to \min_I V$ and $v \to 0$, hence (6.13). Finally

$$t^{\beta}E_N = (t^{\alpha}J_N)^{1/\theta}V(s), \qquad t^{mb}p = v + t^{mb}h\,yu = v + O(t^{-b}),$$

which proves (6.14). A unique minimum gives point convergence by compactness and $y \to 0$. ∎

## 6.3 The degenerating frequency and the remaining positivity issue

For the frozen oscillator $u^2/2 + Ay^{2m} = E_N$, rescaling its period integral yields

$$P = C_m A^{-1/(2m)} E_N^{-\eta}.$$

In the positive-action regime proved above, the normal frequency therefore satisfies

$$\omega \asymp t^{-(m-1)b}, \qquad (t\omega)^{-1} = O(t^{-q}), \qquad |p|/\omega = O(t^{-b}). \tag{6.27}$$

The frequency tends to zero, but the damping and coefficient-variation rates tend to zero faster. These facts are consequences of the action bootstrap, rather than assumptions justifying an informal average. Classical action corrections and nonlinear WKB provide relevant background [6, 10]; neither is invoked as a black box to establish strict positivity in this coupled problem.

Unlike the quadratic case, the ratio $|yu|/E_N$ can grow as $E_N^{-\eta}$. The correction in (6.16) need not be small on an arbitrary very-low-energy tail. Thus nontrivial normal

motion alone has not been shown here to imply a positive rescaled-action limit. The exact transformation

$$y = t^{-b} z(\tau), \qquad \tau = t^q/q$$

gives

$$z_{\tau\tau} + 2ma(s)z^{2m-1} + \frac{b(1-mb)}{q^2}\frac{z}{\tau^2} = 0. \tag{6.28}$$

Its inverse-square coefficient is nonnegative in the selection range. This identity by itself does not remove the preparation hypothesis when $s$ is coupled to $z$.

## 6.4 Actual nonselection above the selection threshold

**Proposition 6.3** (Confined nonselecting trajectories)**.** *For every*

$$\frac{m+1}{m} < \alpha < \frac{m+1}{m-1},$$

*the convex local valley* (6.4) *admits nontrivial-normal solutions of* (6.1)*, with both velocities zero at a sufficiently late positive time, which remain compactly confined and converge to a point* $(s_\infty, 0)$ *with* $a(s_\infty) > \min_I a$*. A neighborhood of such data has the same nonselection property.*

**Proof.** Choose $s_0 = 1/2$ and a compact interval $J \Subset (0,1)$ with $s_0$ in its interior. At time $T$ prescribe

$$s(T) = s_0, \quad p(T) = 0, \quad y(T) = T^{-b}, \quad u(T) = 0.$$

The normal data are nonzero and $T^\beta E(T) = a(s_0)$ is fixed. Until a possible first exit from $J$, Proposition 6.1 gives $|p(t)| + |u(t)| \le Ct^{-mb}$, with $C$ independent of large $T$. Since $mb > 1$,

$$|s(t) - s_0| \le \frac{C}{mb-1} T^{1-mb}.$$

For large $T$ this is smaller than half the distance from $s_0$ to $\partial J$. A first exit is impossible. Bounded energy and compact tangential confinement allow continuation for all $t \ge T$. Finite length gives a limit in $J \times \{0\}$; since $0 \notin J$, that limit does not minimize $a$. At this fixed large time, sufficiently small perturbations of the initial data preserve the uniform scaled-energy bound, nonzero normal data and the strict displacement margin. This proves the neighborhood assertion. ∎

This establishes the boundary of the selection range for the local prepared-data problem. The late positive-time zero-velocity data are not regular cold data at $t = 0$. The coefficient in (6.4) diverges at the ends of $I$, and no global $C^{1,1}$ convex extension is asserted.

## 6.5 A confined open family in the selection range

The preparation theorem is not restricted to the invariant orbit $s = 0$.

**Proposition 6.4** (Nonvacuity of prepared confinement)**.** *For* (6.4) *and each* $1 < \alpha \leq (m+1)/m$*, there are, at sufficiently late fixed times* $T$*, open sets of prepared initial data with nontrivial normal motion whose trajectories stay in a prescribed interval* $J = [-r_0, r_0]$*,* $0 < r_0 < 1$*, and select* $(0,0)$*. Such sets include data with nonzero tangential position and also contain data with both velocities zero at time* $T$*.*

**Proof.** Let the scaled normal data $(T^b y(T), T^{mb} u(T))$ range in a fixed compact neighborhood of $(1,0)$, bounded away from the origin. Let $s(T)$ and $T^{mb} p(T)$ be sufficiently close to zero. These data have uniform upper scaled energy and lower scaled action. All estimates in the proof of Theorem 6.2 apply on the stopped interval before a possible first exit from $J$. The same bootstrap gives a uniform positive lower bound $B_*$ for $B$ there, independently of future confinement. Moreover, the total variation of $B$ and the two reduced $L^1$ error integrals tend uniformly to zero as $T \to \infty$, by (6.20), (6.25) and the displayed integral estimates. Their bounds are constant multiples of sums of $T^{-q}$, $T^{-b}$ and $T^{1-\alpha}$.

Set $U = V - V(0)$. On the stopped interval the exact reduced energy identity is

$$\frac{\mathrm{d}}{\mathrm{d}\sigma}\left(\tfrac{1}{2}v^2 + BU(s)\right) = -\delta v^2 + v\rho + BV'e + \dot{B}U.$$

All nondissipative terms have a total absolute integral tending uniformly to zero; the other factors are uniformly bounded by the full-energy estimate. The boundary barrier is $B_* U(r_0) > 0$. Choose the tangential initial neighborhood so that the initial reduced energy is less than one quarter of this barrier. The correction $t^{mb} h\, yu = O(T^{-b})$ allows this uniformly for large $T$. Increase $T$ so that the total positive error is less than another quarter of the barrier. At a first exit the reduced energy would be at least the full barrier, a contradiction. There is no exit, and Theorem 6.2 applies. Taking interiors of the stated neighborhoods gives open sets at each fixed sufficiently late time, including the asserted choices of positions and velocities. ∎

This is a stopped-interval confinement proof for a specific local coefficient. It gives neither general automatic capture nor singular-time cold initialization. For coefficient functions with a flat minimum interval, Theorem 6.2 asserts minimum-value selection only.

# 7 Scope and the tame-convex question P6

The universal threshold in Theorem 1.1 concerns the full smooth convex class. The valley theorems instead identify concrete mechanisms that restore convergence or select

lower transverse curvature under their stated confinement and action hypotheses. A positive theorem for a smaller class would not change the exact universal boundary $\alpha = 3$.

The next question, denoted P6, is whether every regular cold trajectory converges for every finite-dimensional, globally defined convex $C^2$ semialgebraic objective with globally Lipschitz gradient and nonempty minimizer set, throughout $1 < \alpha < 3$. Polynomially bounded definable objectives of finite regularity, and objectives definable in arbitrary o-minimal expansions, are distinct broader questions. None is settled by the local results of this paper. Semialgebraicity of the minimizer set alone is insufficient: the compact segment in Theorem 2.2 is already semialgebraic.

Several structural cases are simpler and help locate the remaining issue. We give the short arguments to distinguish global hypotheses from local valley assumptions.

**Proposition 7.1** (Global structural reductions)**.** *For a finite-dimensional globally finite differentiable convex objective with globally Lipschitz gradient and nonempty minimizer set, regular cold trajectories converge for every $\alpha > 0$ if the minimizer is unique or if the entire minimizer set is affine. In the affine case the limit is the Euclidean projection of the initial point onto that set. The affine case includes globally real-analytic convex objectives, globally $C^\infty$ convex objectives definable in a polynomially bounded o-minimal expansion, and objectives whose entire minimizer set is an embedded $C^1$ manifold without boundary.*

**Proof.** If $z$ is the unique minimizer, compactness of the unit sphere gives

$$c = \min_{|u|=1} \big(F(z+u) - F(z)\big) > 0.$$

For $r \geq 1$, convexity gives $F(z+ru) - F(z) \geq rc$. Thus $F$ is coercive. Energy bounds the trajectory, and objective convergence for every $\alpha > 0$ [1, Theorem 2.1], applied on any positive-time tail, makes every cluster point equal to $z$.

If $F$ is minimal on a line $z + \mathbb{R}v$, convexity at an arbitrary point $x$ gives

$$\min F \geq F(x) + \langle \nabla F(x), z - x\rangle + t\langle \nabla F(x), v\rangle \qquad (t \in \mathbb{R}).$$

Hence $\langle \nabla F(x), v\rangle = 0$ for all $x$, and $F$ is invariant under translations in direction $v$. For an affine minimizer set $z + L$, the force lies in $L^\perp$, the cold tangential coordinate is constant by (1.5), and the restriction to $z + L^\perp$ has a unique minimizer. Apply the preceding argument to this normal quotient.

If $F$ is globally real analytic, its restriction to the complete line through two distinct minimizers is constant on a segment, hence on the entire line. The translation invariance just proved, applied to all differences of minimizers, makes the minimizer set its affine hull. For the smooth polynomially bounded definable case, the same line restriction cannot cease to be minimal at a finite endpoint: that would give a definable smooth function $h$, flat at zero and positive immediately to its right. Taylor's theorem would imply

$h(t) = o(t^N)$ for every $N$, so $1/h(1/x)$ would grow faster than every power, contradicting polynomial boundedness. Again every minimizing segment extends to a whole minimizing line.

Finally, a nonempty closed convex set that is an embedded $C^1$ manifold without boundary is affine. At a relative-interior point its tangent dimension equals the dimension of its affine hull. Convexity makes it connected, so that dimension is the same everywhere. The inverse function theorem makes the set open in its affine hull; it is also closed there and must equal it. The minimizer set of $F$ is closed, so this argument applies. ∎

Global real analyticity in Proposition 7.1 cannot be replaced by analyticity only on a proper valley chart. Smooth definability in a polynomially bounded structure likewise cannot be replaced by finite differentiability, or by arbitrary o-minimal definability. The absence of smooth flat transitions in the polynomially bounded setting is consistent with the general results in [11]; it does not assert that all such functions are analytic. The genuine finite-regularity target allows nonaffine minimizer sets with boundary or nonsmooth strata.

A universal finite-length route to P6 is already impossible. For $F(x) = x^2/2$ and $\alpha = 2$, the regular cold solution with $x(0) = 1$ is

$$x(t) = \frac{\sin t}{t}, \qquad x'(0) = 0.$$

It converges to zero, but at $t_n = \pi/2 + n\pi$ its values alternate with magnitude $1/t_n$. The sum of the successive absolute increments is $\sum_n (1/t_n + 1/t_{n+1}) = \infty$. Thus even an analytic semialgebraic quadratic can have a point-convergent cold trajectory of infinite length.

The standard definable Kurdyka–Łojasiewicz theory [12] and convergence results with constant positive damping, such as [13], must therefore be used with their actual hypotheses; they do not automatically control tangential drift for damping $\alpha/t$. Geometric rate theorems such as [14, 15] impose additional growth, flatness and damping conditions. The earlier subcritical argument withdrawn in [2, Section 4.3] is not used here. The general finite-regularity tame-convex question remains unresolved here.

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# Appendix A The complete global construction

The global construction below proves bounded cold nonconvergence for each fixed $1 < \alpha < 3$. Its ingredients are cold dependence, elastic reflection with grazing, normal release, convex completion, actual-tail insertion and a static diagonal. Section 2 extends its finite steering step to arbitrary interior targets and strengthens the resulting two distinct minimizing cluster points to the full minimizer segment. Statements and equations in this appendix use their own numbering.

# Smooth convex Nesterov flows without point convergence for damping strictly between one and three

**Abstract**

For each fixed damping parameter $1 < \alpha < 3$, we construct a convex function on $\mathbb{R}^2$ that is globally $C^\infty$, has bounded derivatives of every positive order, and has a compact line segment as its exact minimizer set. The solution of the Nesterov equation from position $(0, 1)$ and zero velocity at the singular time is bounded but does not converge. Its coordinate along the minimizer segment has two distinct subsequential limits. The construction combines finite steering by a nearly linear normal oscillator with exact normal free tails. An elastic wall limit, including grazing contacts, allows all smoothing parameters to be fixed before the height of a new perturbation is decreased. An actual-tail release argument then inserts each perturbation into the original cold trajectory. Summable derivative and finite-prefix budgets produce one static smooth objective. All auxiliary estimates and continuation arguments are proved here. A separate low-damping example and the known convergence implication at and above three give the universal damping threshold in the same smooth class. The objective may depend on $\alpha$, and no discrete-time counterexample is asserted.

**Keywords:** Nesterov flow; convex optimization; point convergence; singular initial data; elastic reflection; smooth convex construction

**MSC2020:** 34D05; 34A12; 90C25

# 1 The result and its context

For a differentiable convex function $F : \mathbb{R}^2 \to \mathbb{R}$, consider

$$X''(t) + \frac{\alpha}{t}X'(t) + \nabla F(X(t)) = 0 \quad (t > 0), \qquad X(0) = (0, 1), \qquad X'(0) = 0. \tag{1.1}$$

A *regular cold solution* means a function $X \in C^2([0, \infty); \mathbb{R}^2)$ satisfying these conditions. In particular, the initial data are imposed at zero. Write $X = (s, y)$, with $s$ the coordinate parallel to the eventual minimizer segment.

**Theorem 1.** *For every fixed $\alpha \in (1, 3)$ there exist $S > d > 0$ and a nonnegative convex function $F = F_\alpha \in C^\infty(\mathbb{R}^2)$ such that*

$$\|D^j F\|_{L^\infty(\mathbb{R}^2)} < \infty \quad (j \geq 1), \qquad \operatorname{argmin} F = [-S, S] \times \{0\}. \tag{1.2}$$

*Its unique regular cold solution of (1.1) is bounded. There are times $t_n \to \infty$ for which*

$$s(t_{2n}) \longrightarrow d, \qquad s(t_{2n+1}) \longrightarrow -d. \tag{1.3}$$

*Thus this solution does not converge. Its potential and mechanical energies nevertheless tend to zero, and $y(t) \to 0$.*

The norm of $D^jF$ in (1.2) is the operator norm of the $j$-linear derivative; any fixed equivalent tensor norm would give the same assertion. In particular, the gradient is globally bounded and globally Lipschitz. Boundedness of $F$ itself is neither required nor asserted.

There are three relevant distinctions in the literature. Attouch, Chbani and Riahi prove the subcritical objective estimate $F(X(t)) - \min F = O(t^{-2\alpha/3})$ in Theorem 2.1 of [1]; their Theorem 3.2 proves scalar point convergence for $\alpha = 3$. Objective convergence is compatible with the conclusion of Theorem 1. Jang and Ryu prove finite-dimensional point convergence for the critical Nesterov equation in Theorem 2 of [2]. Section 4.3 of its second version, dated 19 January 2026, withdraws the earlier subcritical argument and explicitly leaves the range $[1, 3)$ open in that paper. We use neither the withdrawn argument nor an extrapolation of the critical theorem. Ryu's Theorems 2.1 and 3.3 in [3] concern a critical cold trajectory for a convex $C^1$ function on $\mathbb{R}^2$, with infinite length and divergent time-weighted potential and kinetic integrals; that trajectory converges to its unique minimizer. These are different conclusions from point nonconvergence. These comparisons specify the scope of the cited statements, rather than a historical priority claim.

The proof has an order of choices that is important. A normal oscillator with a narrow region of zero force activates a weak tangential well for most of a long finite interval. This steers the tangential coordinate near a chosen target. Elastic walls bound its later motion while a normal core radius is selected by a zero-count argument. A symmetric convex completion preserves exact inner and outer product regions. Only after this entire normalized shape is fixed do we lower its physical height. The fixed physical wall then becomes a diverging penalty in normalized time. We prove convergence to elastic reflection, including at zero-speed contacts, and separately prove that the actual old trajectory reaches a bounded-energy normalized state. Finally, a static series preserves alternating finite observations. No finite-stage eventual tangential endpoint is required to be near the selected target.

# 2 Cold equations and scalar identities

Throughout the construction fix $1 < \alpha < 3$ and put

$$a = \alpha - 1, \qquad \beta = \frac{2\alpha}{3}. \tag{2.1}$$

The wall-limit lemma below also applies for $\alpha > 1$.

**Lemma 1** (Cold existence and dependence)**.** *For a globally Lipschitz force $G : \mathbb{R}^k \to \mathbb{R}^k$ and initial position $x_0$, there is a unique global regular cold solution. It is equivalent to*

$$x'(t) = -t^{-\alpha} \int_0^t v^\alpha G(x(v))\, \mathrm{d}v, \tag{2.2}$$

$$x(t) = x_0 - \frac{1}{a}\int_0^t v\left[1 - \left(\frac{v}{t}\right)^a\right] G(x(v))\,\mathrm{d}v. \tag{2.3}$$

*It satisfies* $x''(0) = -G(x_0)/(\alpha+1)$. *If two forces have Lipschitz constants at most* $L_0$, *have sup-norm difference at most* $h$, *and use the same cold data, then on* $[0,T]$

$$\|x - \widetilde{x}\|_\infty \le B_T(h) := \frac{hT^2}{2a}\exp\!\left(\frac{L_0T^2}{2a}\right), \tag{2.4}$$

$$\|x' - \widetilde{x}'\|_\infty \le \frac{T}{\alpha+1}\big(h + L_0B_T(h)\big). \tag{2.5}$$

*The corresponding local conclusions hold for locally Lipschitz forces as long as the trajectories remain in a fixed compact set.*

**Proof.** Integrating $(t^\alpha x')' = -t^\alpha G(x)$ and reversing the order of integration gives the two formulas. The operator on the right of (2.3) is a contraction on a suitable closed ball of continuous paths for small $T$, since its Lipschitz constant is at most $L_0T^2/(2a)$. Differentiation gives (2.2) and $x'(t) = -G(x_0)t/(\alpha+1) + o(t)$. Substitution in the equation proves continuity of $x''$ at zero and the asserted second derivative. Conversely every regular solution integrates to these formulas. Positive-time continuation is global for a globally Lipschitz force, which has at most linear growth.

For the difference let $D(t) = \sup_{0\le v\le t}|x(v) - \widetilde{x}(v)|$. The nonnegative Volterra kernel is bounded by $v/a$, so

$$D(t) \le \frac{ht^2}{2a} + \frac{L_0}{a}\int_0^t vD(v)\,\mathrm{d}v.$$

Integral Gronwall gives (2.4); the velocity formula gives (2.5). Local versions follow by a first-exit argument. For local uniform force convergence to a fixed locally Lipschitz force, split the difference using that limiting force's Lipschitz constant. No uniform Lipschitz bound on the approximating forces is needed for this local version. ∎

For a scalar potential $V \ge 0$ the equation $x'' + \alpha x'/t + V'(x) = 0$ gives

$$e = \tfrac{1}{2}x'^2 + V(x), \qquad e' = -\frac{\alpha}{t}x'^2, \tag{2.6}$$

$$q = x + \frac{tx'}{a}, \qquad q' = -\frac{t}{a}V'(x), \tag{2.7}$$

$$M = \tfrac{1}{2}(tx' + ax)^2 + t^2V(x), \qquad M' = t\big((3-\alpha)V - a(xV' - V)\big). \tag{2.8}$$

These identities follow by differentiation, using $(tx' + ax)' = -tV'(x)$. They hold almost everywhere for the piecewise smooth scalar forces used below, and their energy quantities are continuous at transverse switches. On a free leg from time $T > 0$,

$$x(t) = q + [x(T) - q](T/t)^a, \qquad q = x(T) + Tx'(T)/a. \tag{2.9}$$

**Lemma 2** (The sign-force reference)**.** *The piecewise classical cold solution*

$$u'' + \frac{\alpha}{t}u' + \operatorname{sgn} u = 0, \qquad u(0) = 1, \quad u'(0) = 0 \tag{2.10}$$

*is global and bounded and has infinitely many transverse zeros with no finite accumulation. If* $e = u'^2/2 + |u|$*, then*

$$t^\beta e(t) \longrightarrow C_\alpha \in (0, \infty). \tag{2.11}$$

**Proof.** The first positive leg is $u = 1 - t^2/[2(\alpha+1)]$. Continue at each zero with continuous velocity. Since $e' = -\alpha u'^2/t \geq -2\alpha e/t$, the energy is strictly positive at every positive finite time and at most one. Zeros are transverse. A finite accumulation would force the continuous velocity to vanish at a zero, whereas the positive lower energy bound prevents this. The acceleration is bounded away from time zero and the energy bounds both position and velocity, so the continuation is global. If the solution were eventually positive, integration of $(t^\alpha u')' = -t^\alpha$ would give $u' = -t/(\alpha+1) + O(t^{-\alpha})$, forcing a crossing. Reflection excludes an eventual negative sign too.

Set $H = e + \beta u u'/t$. Direct differentiation gives

$$H' + \frac{\beta}{t} H = -\frac{2\alpha(\alpha+3)}{9t^2} u u'. \tag{2.12}$$

Indeed $(uu')' = u'^2 - \alpha u u'/t - |u|$, and the coefficients of $u'^2$ and $|u|$ cancel for $\beta = 2\alpha/3$. Since $|u| \leq e$ and $|u'| \leq \sqrt{2e}$, $|uu'| \leq \sqrt{2} e^{3/2}$ and $H/e \to 1$. For large time $H > 0$ and

$$\left| (\log H)' + \frac{\beta}{t} \right| \leq C_\alpha t^{-2}.$$

Thus $\log(t^\beta H)$ has a finite real limit. Exponentiation and $H/e \to 1$ prove (2.11). This argument uses neither a period approximation nor an averaged equation. ∎

**Lemma 3** (A bounded harmonic propagator)**.** *Let* $c = c_\alpha$ *solve*

$$c'' + \frac{\alpha}{\tau} c' + c = 0, \qquad c(0) = 1, \quad c'(0) = 0. \tag{2.13}$$

*Then* $|c| + |c'| \leq C\tau^{-\alpha/2}$ *for* $\tau \geq 1$*,* $c(\tau) < 1$ *for* $\tau > 0$*, and*

$$\rho_\alpha := -\inf_{\tau \geq 0} c(\tau) \in [0, 1). \tag{2.14}$$

*Suppose* $0 \leq d \leq 1$ *and* $z'' + \alpha z'/\tau + z = d(\tau) z$ *has cold data* $(z_0, 0)$ *on* $[0, L]$*, where* $\int_0^L d \leq 1$*. Then*

$$|z(\tau)| + |z'(\tau)| \leq C|z_0|\tau^{-\alpha/2} \qquad (1 \leq \tau \leq L), \tag{2.15}$$

$$\sup_{0 \leq \tau \leq L} |z(\tau) - z_0 c(\tau)| \leq C|z_0| \int_0^L d(\tau)\, \mathrm{d}\tau. \tag{2.16}$$

*Constants depend only on* $\alpha$*.*

**Proof.** The preliminary energy $J = z^2 + z'^2$ satisfies $J' = -2\alpha z'^2/\tau + 2dzz' \leq dJ$, so $J \leq z_0^2 \exp(\int d)$. For $\tau \geq 1$ set $w = \tau^{\alpha/2} z$. Then

$$w'' + \left(1 + \frac{\alpha(2-\alpha)}{4\tau^2}\right) w = dw. \tag{2.17}$$

Its energy $K = w^2 + w'^2$ obeys $|K'| \le (C_\alpha \tau^{-2} + d)K$. When $L \ge 1$, the data at one are bounded by the preliminary estimate. Integrability of $\tau^{-2}$ and the bound on $\int d$ prove (2.15). With $d = 0$ this proves the decay of $c$. Its ordinary oscillator energy is strictly less than its initial value $1/2$ at every positive time, since $c'$ is nonzero immediately after zero and dissipation is positive on a nonempty interval. Hence $|c(\tau)| < 1$ for $\tau > 0$. Its decay, continuity, and positivity near zero imply (2.14).

For completeness the homogeneous first-order propagator associated with (2.17) at $d = 0$ is bounded uniformly between any two times $1 \le \sigma \le \tau$: the preceding energy inequality bounds its norm by $\exp(C \int_\sigma^\tau v^{-2}\,\mathrm{d}v)$. Let $e = z - z_0 c$. Its equation is $e'' + \alpha e'/\tau + e = dz$ with zero cold data. Put $L_1 = \min(1, L)$. On $[0, L_1]$, the cold Volterra formula for this equation, together with the preliminary bound on $z$, gives

$$\sup_{0 \le \tau \le L_1} (|e(\tau)| + |e'(\tau)|) \le C|z_0| \int_0^{L_1} d(\tau)\,\mathrm{d}\tau.$$

If $L \le 1$, this proves the required position comparison. The decay assertion is empty for $L < 1$ and follows from the preliminary bound at $L = 1$. If $L > 1$, these bounds provide the data at one. For $\tau \ge 1$ the conjugated variable $v = \tau^{\alpha/2} e$ has forcing $\tau^{\alpha/2} dz$, whose absolute integral is at most $C|z_0| \int_1^L d$ by (2.15). Variation of constants with the bounded propagator proves $|v| + |v'| \le C|z_0| \int_0^L d$. This proves (2.16) with a constant independent of $L$. ∎

# 3 The elastic wall limit including grazing

We use fixed positive diagonal masses $M_s, M_u$ for coordinates $(s, u)$. The generality is limited to diagonal masses: a tangential impulse would in general change both velocities for a mass matrix with an off-diagonal term.

**Lemma 4** (Finite-prefix wall limit)**.** *Let $\alpha > 1$, $S, \eta > 0$, and $m \in (-S, S)$. Suppose $V \ge 0$ is $C^2$ on the strip $|s| < S + \eta$ and satisfies*

$$(s - m) V_s(s, u) \ge 0. \tag{3.1}$$

*Let $W \in C^2(\mathbb{R})$ be convex, coercive, zero on $[-S, S]$, and strictly positive outside. Consider*

$$M_s(s_\varepsilon'' + \alpha s_\varepsilon'/t) + V_s(s_\varepsilon, u_\varepsilon) + \varepsilon^{-1} W'(s_\varepsilon) = 0, \tag{3.2}$$

$$M_u(u_\varepsilon'' + \alpha u_\varepsilon'/t) + V_u(s_\varepsilon, u_\varepsilon) = 0, \tag{3.3}$$

*with cold data $(s_0, u_0)$, $|s_0| \le S$. On every fixed finite interval these solutions converge, as $\varepsilon \downarrow 0$, to a unique elastic trajectory confined to $[-S, S]$ in its first coordinate. In the interior it solves the equations without $W$. At a nonzero-speed wall contact $s'$ changes to $-s'$, while $u, u'$ remain continuous. A zero-speed contact has the unique ordinary*

*ODE continuation, possibly with an interval at rest on the wall. There are finitely many connected wall-contact episodes on every bounded time interval.*

*Positions and $u'$ converge uniformly on finite intervals. The velocities $s'_\varepsilon$ converge in every finite $L^p$ norm, and uniformly on compact sets avoiding the limiting nonzero impacts, including through grazing and resting contacts. Mechanical energies including the penalty converge uniformly. The same topology holds for hard-wall solutions under local $C^1$ changes of the smooth nonnegative potentials, provided every potential satisfies* (3.1) *for the same $m$ and the cold data converge within the wall interval. It also holds for penalized solutions starting at a fixed positive time from data converging to a strictly interior state, with both velocities converging and with zero initial wall potential.*

**Proof.** We give the argument for $M_s = M_u = 1$ and specify the mass changes at the end. To prove a statement on $[0, T]$, choose an auxiliary horizon $H > T$ and carry out all trace and measure arguments before restricting to $[0, T]$.

*Energy and the initial interval.* The energy identity is

$$E_\varepsilon = \tfrac{1}{2}(s_\varepsilon'^2 + u_\varepsilon'^2) + V(s_\varepsilon, u_\varepsilon) + \varepsilon^{-1} W(s_\varepsilon), \qquad E'_\varepsilon = -\frac{\alpha}{t}(s_\varepsilon'^2 + u_\varepsilon'^2). \tag{3.4}$$

It extends to zero by the cold formulas. Thus $E_\varepsilon \le E_0 = V(s_0, u_0)$ and all speeds are at most $B = \sqrt{2E_0}$. If $E_0 = 0$ the solution is stationary, since a nonnegative differentiable function has zero gradient at a zero. Otherwise the wall values at $\pm(S+\eta/2)$ exclude strip exit for small $\varepsilon$, and $|u_\varepsilon| \le |u_0| + BH$. These estimates also give continuation through $H$.

An interior start has a common short wall-free interval. At $s_0 = S$ the speed bound keeps $s_\varepsilon > m$ on a common interval, where

$$(t^\alpha s'_\varepsilon)' = -t^\alpha (V_s + \varepsilon^{-1} W') \le 0.$$

Starting with zero velocity, the solution cannot move outside $S$. Reflection gives the left-boundary case. Hence on a common $[0, t_*]$ all solutions coincide with the ordinary cold solution without the penalty. The local Volterra uniqueness in Lemma 1 justifies this assertion at the singular origin.

*Excursion count and compactness.* Let

$$\delta_\varepsilon = \sup\{\operatorname{dist}(s, [-S, S]) : W(s) \le \varepsilon E_0\}.$$

Coercivity and strict positivity outside the interval imply $\delta_\varepsilon \to 0$. All positions lie between $-S - \delta_\varepsilon$ and $S + \delta_\varepsilon$. On a connected right exterior excursion $s_\varepsilon > S$, its weighted velocity is strictly decreasing, since convexity gives $W'(s) > 0$ for $s > S$. There is at most one turn and a completed excursion returns with negative velocity. Until it subsequently passes $m$, the weighted velocity remains nonincreasing and it cannot begin another exterior excursion. The left side is symmetric. Writing $d_* = \min(S - m, S + m) > 0$, consecutive excursions require at least $d_*$ of tangential travel. Since total travel on $[0, H]$ is at most

$BH$, their number is at most $2 + BH/d_*$. This bound also counts an unfinished last excursion.

On $[t_*, H]$ ordinary forces are uniformly bounded. The normal accelerations are bounded, while the positions are uniformly Lipschitz. On one exterior excursion, possibly truncated,

$$\int \varepsilon^{-1}|W'(s_\varepsilon)|\,\mathrm{d}t \le |s'_\varepsilon(b) - s'_\varepsilon(a)| + \int \left(\frac{\alpha}{t}|s'_\varepsilon| + |V_s|\right)\,\mathrm{d}t.$$

The finite excursion count bounds the sum uniformly. Thus the wall force has bounded $L^1$ norm and $s'_\varepsilon$ has bounded variation. We may extract uniform limits of positions and $u'$, an $L^1$ and almost everywhere limit $v$ of $s'_\varepsilon$, and a weak measure limit $\mu$ of the signed wall forces. The position limit has derivative $v$ almost everywhere. Uniform boundedness upgrades $L^1$ convergence to every finite $L^p$.

*Elasticity from energy.* Convexity of $W$ gives

$$0 \le W(s) \le \operatorname{dist}(s, [-S, S])|W'(s)|.$$

Consequently

$$\int_{t_*}^{H} \varepsilon^{-1} W(s_\varepsilon)\,\mathrm{d}t \longrightarrow 0. \tag{3.5}$$

The energies in (3.4) are equi-Lipschitz away from zero. Pass to a uniformly convergent subsequence with continuous limit $\overline{E}$. The convergences just proved and (3.5) give, almost everywhere,

$$\overline{E} = \tfrac{1}{2}(v^2 + u'^2) + V(s, u). \tag{3.6}$$

At each interior time of the enlarged interval, approach from both sides through this full-measure set. A bounded-variation function has one-sided traces. Continuity of the other terms yields

$$v(t-)^2 = v(t+)^2. \tag{3.7}$$

This proves conservation of speed at any possible jump; uniform position convergence alone would not prove it. No pointwise vanishing of penalty energy at an impact is asserted.

The limiting equations are

$$\mathrm{d}v + \left(\frac{\alpha}{t}v + V_s(s, u)\right)\,\mathrm{d}t + \mathrm{d}\mu = 0, \qquad u'' + \frac{\alpha}{t}u' + V_u(s, u) = 0. \tag{3.8}$$

The measure is supported on $s = \pm S$, since a compact subset of the interior is eventually wall-free. It is nonnegative at $S$ and nonpositive at $-S$. There is no reaction near zero.

*Contacts, including grazing.* On a connected component $J$ of $\{s > m\}$, the weighted velocity $p = t^\alpha v$ is nonincreasing as a distribution, by (3.8). Thus $s$ first increases, may remain constant, and then decreases on $J$. The set where $s = S$ in this component is empty, a singleton, or one interval. On the interior of a contact interval $v = 0$, and

$$0 = \mathrm{d}p = -t^\alpha V_s(S, u)\,\mathrm{d}t - t^\alpha \mathrm{d}\mu$$

is a sum of two nonpositive measures. Both vanish. In particular $V_s(S,u) = 0$ there by continuity, and no diffuse reaction remains. Only the endpoints of the interval could carry atoms. At an endpoint adjoining rest, one trace of $v$ is zero, so (3.7) makes the other zero and the atom vanishes. At a singleton contact confinement gives $v_- \geq 0$, $v_+ \leq 0$. Together with (3.7) and $v_+ - v_- = -\mu(\{t\})$, this gives $v_+ = -v_-$. The left wall has the reflected argument. There is no residual continuous singular reaction. Different contact episodes require a visit to $m$ between them, and the bounded travel gives finitely many on the horizon.

At a zero-speed contact at positive time, the ordinary locally Lipschitz ODE has a unique continuation. Its weighted velocity is nonincreasing from zero while $s > m$ at the right wall, so this continuation cannot leave through that wall. A constrained solution cannot stay at rest when the inward force is positive, by the measure calculation, and cannot jump spontaneously from rest by (3.7). Hence it is this same ordinary continuation. The conclusion covers a boundary rest interval regardless of how many times the smooth boundary force vanishes or activates during that interval.

Starting with the common initial segment, ordinary uniqueness, reflection at nonzero hits, and the just-proved grazing uniqueness identify one trajectory. The finite contact count prevents finite accumulation of impacts. Every compactness subsequence has this same limit, proving full-family convergence. Its energy is continuous across elastic impacts, so (3.6) identifies the uniform energy limit at all times using its continuous mechanical interpretation.

*The velocity topology.* In an interior time neighborhood the wall term eventually vanishes and tangential accelerations are bounded, upgrading $L^1$ to uniform velocity convergence on a smaller neighborhood. Near a zero-speed right contact choose a neighborhood in $s > m$ avoiding nonzero impacts. The functions $t^\alpha s'_\varepsilon$ are monotone there and converge almost everywhere to a continuous function. This implies uniform convergence on smaller compact intervals: sandwich values between neighboring good points and use a finite partition and uniform continuity of the limit. Reflection handles the other wall. The enlarged horizon provides these neighborhoods at the target endpoint $T$, and the common initial segment handles zero. A finite covering proves the asserted uniform topology. Uniform convergence across a nonzero impact is impossible and is not used.

*Masses and dependence.* For fixed positive diagonal masses, use kinetic energy $(M_s s'^2 + M_u u'^2)/2$, weighted momentum $M_s t^\alpha s'$, and jump equation $M_s(v_+ - v_-) = -\mu(\{t\})$. All bounds change by fixed constants, and (3.7) and the same reflection law follow.

For hard trajectories of nonnegative potentials $V_n \to V$ in $C^1_{\mathrm{loc}}$ satisfying the same inward-sign condition, initial energies and compact force bounds are uniform. Converging cold positions in the wall interval have a common initial wall-free segment by the same inward argument; Volterra dependence replaces literal equality on it. On the rest of the interval the travel count bounds the number and size of impulses, giving the same

BV compactness. Energies now equal ordinary mechanical energy exactly, so the energy, measure, contact, and uniqueness arguments apply without (3.5). This proves shape dependence, including through grazing. Finally, for convergent data at a positive interior time, zero wall potential and convergent velocities bound the initial energies and provide a short wall-free interval. The proof then starts at that positive time. No initial boundary-velocity convention is required in this version. ∎

# 4 Convex blocks and sharp normal interfaces

Fix $S > 0$ and a compact tangential guard $I = [-S - \eta, S + \eta]$ with $\eta > 0$. We will use targets $m$ in a fixed small central interval. Choose $M$ so large that the symmetric construction strip

$$D_m = \{(s, u) : |s - m| < M\}$$

contains $I \times \mathbb{R}$ with a fixed positive tangential margin, and also $M > e^{1/2}(S + |m|) + 1$ for all the selected targets. All constants in this section may depend on this geometry and on $\alpha$.

Choose an even smooth function $\theta$ between zero and one, nondecreasing on $[0, \infty)$, equal to zero for $|v| \le 2$ and to one for $|v| \ge 3$. Its transition is confined to a compact subinterval of $(2, 3)$ and its reflection. Set $\psi = \theta^2$. Fix $0 < \gamma < 1/2$. For sufficiently small $\kappa > 0$, there is an even nonnegative smooth function $b$ supported in $2 < |v| < 3$ with

$$\int_2^3 b(v)\,\mathrm{d}v = \gamma, \qquad b \ge \kappa M^2(6\theta'^2 - 2\theta\theta'')_+. \tag{4.1}$$

To construct it, take a smooth positive majorant of the indicated compactly supported function on a slightly larger compact transition interval, normalize its integral, and then decrease $\kappa$. Let $P$ be even, with $P = P' = 0$ on $[-2, 2]$ and $P'' = b$, and put $c_* = 1 - \gamma$. For $0 \le \ell < r < 1/8$, define

$$g_{r,\ell}(u) = c_*(|u| - \ell)_+ + rP(u/r), \tag{4.2}$$

$$h_r(u) = \kappa r\psi(u/r), \tag{4.3}$$

$$V_{r,\ell}(s, u) = g_{r,\ell}(u) + h_r(u)(s - m)^2. \tag{4.4}$$

The inner normal force is sharp at $\pm\ell$, or at zero if $\ell = 0$.

**Lemma 5** (Convexity and shape bounds)**.** *The function $V_{r,\ell}$ is nonnegative and convex on $D_m$. It has the inward sign* (3.1)*. Uniformly for $0 \le \ell < r$ on this strip,*

$$|g_{r,\ell}(u) - |u|| \le Cr, \qquad |V_{r,\ell} - u(V_{r,\ell})_u| \le Cr, \tag{4.5}$$

$$|(V_{r,\ell})_u| \le C, \qquad |(V_{r,\ell})_s| \le Cr. \tag{4.6}$$

*Derivative statements at sharp points are understood on either smooth side. For $|u| < 2r$ the potential is exactly $c_*(|u| - \ell)_+$ and independent of $s$. For $|u| > 3r$ it is*

$$|u| - c_*\ell - r\mu_b + \kappa r(s-m)^2, \qquad \mu_b = \int_2^3 vb(v)\,\mathrm{d}v. \tag{4.7}$$

*For the elastic-wall cold solution from $(s_0, 1)$, $|s_0| \le S$, with diagonal masses $M_s = A^{-2}$, $M_u = 1$ and fixed $A > 0$,*

$$|u(t)| \le 1 + C_* r \qquad (t \ge 0). \tag{4.8}$$

*Here $C_*$ can be chosen independently of $\ell$ and $s_0$.*

**Proof.** Write $z = s - m$. For $g(u) + h(u)z^2$, the Schur complement of the positive tangential Hessian entry $2h$ is

$$g'' + \left(h'' - \frac{2h'^2}{h}\right) z^2.$$

For $h = \kappa r\theta(u/r)^2$, its coefficient of $z^2$ equals $-(\kappa/r)(6\theta'^2 - 2\theta\theta'')(u/r)$. The padding (4.1) therefore makes the Hessian nonnegative wherever $h > 0$. At its zeros the tangential and mixed entries vanish and the remaining normal curvature is nonnegative. The inner hinge supplies a nonnegative normal curvature measure in a region where $h = 0$, so convexity holds across the sharp interfaces. The tangential derivative is $2h_r(s-m)$, proving its sign.

For $v > 3$, $P(v) = \gamma v - \mu_b$, which gives (4.7). On the transition and inner region all the errors in (4.5) are bounded by a constant times $r$; outside, the formulas are exact. Differentiation gives (4.6). The cold energy is at most $1 + C_1 r$ and the potential is at least $|u| - C_2 r$. Elastic reflection preserves energy and damping decreases it. Thus (4.8) follows with $C_* = C_1 + C_2$. Existence of the sharp elastic solution, used in this last sentence, is supplied by the next lemma. ∎

**Lemma 6** (Sharp interfaces and radius dependence)**.** *The family* (4.4) *has unique global elastic cold solutions with the masses just specified. Every finite normal zero and every finite crossing of a sharp normal interface is transverse. There are finitely many such events on a bounded time interval. Under $\ell_j \to \ell \ge 0$, positions and normal velocities converge uniformly on every finite interval, and tangential velocities have the topology of Lemma 4. The same conclusions hold under monotone smoothing of the inner force in shrinking neighborhoods of its interfaces. In particular, the exact number of normal zeros on a finite prefix is locally constant if its endpoint is not a zero.*

**Proof.** Smooth the inner convex hinge by odd nondecreasing normal forces, zero in a slightly smaller core when $\ell > 0$, and equal to $c_* \operatorname{sgn} u$ outside arbitrarily thin strips around $\pm\ell$. At $\ell = 0$ smooth near zero. Normalize the primitive to vanish at zero. These changes have uniformly bounded forces and uniformly vanishing value error, retain the product region $|u| < 2r$, and do not change the tangential force. Lemma 4 applies to

every smoothed potential. Near the cold position $u = 1$ the forces are identical. The inward argument supplies a common initial segment even when $s_0$ is a wall endpoint.

On an enlarged positive-time compact interval, energy bounds both speeds and positions, normal accelerations are bounded, and the inward sign bounds contact count and tangential velocity variation. The proof of Lemma 4 gives subsequential uniform position and normal-velocity limits, $L^1$ tangential-velocity limits, and measure limits of tangential impulses. Since the potentials converge in value, the equi-Lipschitz energies identify elastic tangential traces as before. The unchanged continuous tangential force gives the same contact classification, including grazing.

It remains to identify the sharp normal dynamics. In a fixed neighborhood of an interface, all approximants are exactly scalar in $u$. Their local normal energy satisfies the integrated identity

$$e_j(t_2) - e_j(t_1) = -\alpha \int_{t_1}^{t_2} \frac{u_j'^2}{t}\,\mathrm{d}t.$$

Uniform convergence of position, normal velocity, and potential passes this to the limit with

$$e = \tfrac{1}{2}u'^2 + c_*(|u| - \ell)_+ \quad \text{or} \quad e = \tfrac{1}{2}u'^2 + c_*|u| \text{ if } \ell = 0.$$

Thus $e' = -\alpha u'^2/t \geq -2\alpha e/t$ locally. A zero-velocity arrival at an interface would have $e = 0$. Forward monotonicity and backward integration of this lower differential inequality force $e = 0$ on a neighborhood of that time. The normal coordinate is constant at the interface there. This constant interval extends backward through every positive endpoint by the same local argument, because its coordinate remains in the scalar product region. It would give $u(0) = \pm\ell$ or zero, contradicting $u(0) = 1$. A zero with zero velocity in a positive flat core is excluded in the same way, or by backward uniqueness of the free equation. All finite interface events and zeros are therefore transverse.

A finite accumulation of interfaces or zeros would, by continuity and Rolle's theorem, give a zero-velocity event of precisely the type excluded. Near time zero there are no events. Off the finitely many events the forces converge uniformly along the trajectories, so the limit solves the required sharp equation almost everywhere. The assigned force at an isolated switch does not affect it. Ordinary uniqueness between events and transverse continuation at each event prove uniqueness. At a simultaneous sharp normal switch and tangential contact the coupling is identically zero in a neighborhood; the scalar normal crossing and tangential reflection or free grazing continuation are independent and uniquely defined. Thus there is no unproved smooth-dependence assertion at a joint nonsmooth event. Every compactness subsequence has the same limit.

For changing radii, the potentials converge uniformly, the normal forces converge away from the limiting interfaces, and all nearby scalar patches have the same local energy identity. The same proof applies verbatim. Near a transverse zero the limiting normal velocity keeps a fixed nonzero sign on a small neighborhood; uniform normal-velocity

convergence gives exactly one nearby zero. On the complement of these finitely many neighborhoods the normal position is bounded away from zero. This proves the last assertion. The energy bound also yields global continuation, proving all claims. ∎

**Lemma 7** (Energy decay with elastic confinement)**.** *For each block above, its elastic cold energy tends to zero. The same conclusion holds for a bounded smooth convex trajectory with nonnegative potential having a zero-valued minimizer.*

**Proof.** For the block put

$$K = u'^2 + A^{-2}s'^2, \qquad E = K/2 + V, \qquad b = uu' + A^{-2}(s-m)s'.$$

The coordinates and speeds are bounded, so $b$ is bounded. The continuous energy satisfies $E' = -\alpha K/t$ almost everywhere; hence $\int_1^\infty K/t < \infty$. In mass-normalized measure form

$$A^{-2}\mathrm{d}s' + (\alpha A^{-2}s'/t + V_s)\,\mathrm{d}t + \mathrm{d}\mu = 0.$$

Its virial identity is

$$\mathrm{d}b = [K - \alpha b/t - uV_u - (s-m)V_s]\,\mathrm{d}t - (s-m)\mathrm{d}\mu. \tag{4.9}$$

The last measure is nonpositive: at $S$, both $s-m$ and $\mu$ are nonnegative, and at $-S$ both are nonpositive. Convexity relative to the zero-valued point $(m,0)$ gives $V \le uV_u + (s-m)V_s$ almost everywhere, including along the sharp solution away from its isolated normal switches. Choose a fixed $t_v \ge 1$ away from nonzero tangential impacts. For $T > t_v$ also away from such impacts, divide (4.9) by $t$, discard its favorable reaction term, and integrate by parts:

$$\int_{t_v}^T \frac{V}{t}\,\mathrm{d}t \le \int_{t_v}^T \frac{K}{t}\,\mathrm{d}t - \frac{b(T)}{T} + \frac{b(t_v)}{t_v} - (\alpha+1)\int_{t_v}^T \frac{b}{t^2}\,\mathrm{d}t.$$

The right side is bounded above uniformly in $T$. The bound extends to arbitrary terminal times by limits from nonimpact times, and adding the finite interval $[1, t_v]$ is harmless. Therefore $\int_1^\infty E/t < \infty$. Since $E$ is nonnegative and nonincreasing, $E(t) \to 0$. No global bound on the total wall reaction is needed; its favorable sign is used on each finite interval.

For an ordinary bounded convex trajectory take $b = \langle X - z, X' \rangle$ for a minimizer $z$, $K = |X'|^2$, and omit the reaction measure. Energy bounds velocities and hence $b$. Exactly the same calculation proves the assertion. ∎

For future use put $b_0 = 3r/2$. For every $\ell < r$,

$$V_{r,\ell}(s, \pm b_0) = c_*(b_0 - \ell) > c_* r/2. \tag{4.10}$$

Once $E < c_* r/8$, the trajectory remains strictly in $|u| < b_0$. The coordinates are then exactly decoupled. Such a strict barrier can be imposed at a fixed time away from nonzero tangential impacts, and is preserved under sufficiently small changes of radius or smoothing by the preceding finite-prefix dependence.

# 5 Finite steering before the inner radius is chosen

Choose a target $m$ satisfying

$$(1+\rho_\alpha)|m| < (1-\rho_\alpha)S, \tag{5.1}$$

where $\rho_\alpha$ is given by (2.14). The stronger central condition, rather than merely $|m| < S$, gives a common wall-free steering interval for every starting $s_0 \in [-S, S]$.

**Lemma 8** (Uniform early steering)**.** *Fix $A > 0$, $s_0 \in [-S, S]$, and an admissible target $m$. Choose*

$$\frac{\alpha}{3} < q < \frac{3\alpha}{2\alpha+3}, \qquad T_r = r^{-q/\beta}, \qquad \omega = A\sqrt{2\kappa r}. \tag{5.2}$$

*For all sufficiently small $r$, uniformly over $0 \le \ell < r$, the cold block trajectory from $(s_0, 1)$ has no positive-time wall contact on $[0, T_r]$ and satisfies*

$$|s(T_r) - m| \le C(\omega T_r)^{-\alpha/2} \longrightarrow 0, \tag{5.3}$$

$$T_r|s'(T_r)| \le C(\omega T_r)^{1-\alpha/2}, \tag{5.4}$$

$$|u(T_r)| \le Cr^q. \tag{5.5}$$

*The right side of* (5.4) *tends to zero only when $\alpha > 2$. No vanishing weighted-speed assertion is made for $1 < \alpha \le 2$. The final static construction retains only the position observations in* (5.3)*.*

**Proof.** Initially work with the unconstrained local block equation in $D_m$, up to a possible first strip exit. Its sharp scalar switches are constructed by the local argument of Lemma 6; no wall is used in this part of the proof. Put $z = s - m$ and use

$$u'' + \alpha u'/t + V_u = 0, \qquad z'' + \alpha z'/t + \omega^2 \psi(u/r) z = 0. \tag{5.6}$$

Its actual total energy, which includes energy exchanged through the transition, is

$$E = \tfrac{1}{2}u'^2 + \frac{z'^2}{2A^2} + V, \qquad E' = -\frac{\alpha}{t}(u'^2 + A^{-2}z'^2).$$

In tangential clock $\tau = \omega t$, set

$$Q(t) = \omega \int_0^t [1 - \psi(u(v)/r)]\, \mathrm{d}v, \qquad J = z^2 + z_\tau^2. \tag{5.7}$$

The exact equation gives $J_\tau \le [1 - \psi(u/r)]J$, so

$$J(t) \le z(0)^2 e^{Q(t)}. \tag{5.8}$$

While $Q \le 1$, this retains a strict construction-strip margin by our choice of $M$, and yields $A^{-2}z'^2 \le Cr$. For the actual scalar part $e_N = u'^2/2 + g_{r,\ell}(u)$, this also gives

$$0 \le E - e_N = \frac{z'^2}{2A^2} + h_r(u)z^2 \le Cr.$$

We use this exact comparison, not an autonomous scalar dissipation identity during the coupled transition.

*The coupled energy estimate.* Set $H = E + \beta uu'/t$. Direct differentiation, without discarding any transition terms, gives

$$H' + \frac{\beta}{t}H = \frac{\beta}{t}(V - uV_u - A^{-2}z'^2) - \frac{2\alpha(\alpha+3)}{9t^2}uu'. \tag{5.9}$$

The first bracket is $O(r)$ while $Q \le 1$. If also $E \ge C_0 r$ for a sufficiently large fixed constant $C_0$, then $|u| \le E + Cr \le CE$ and $|uu'| \le CE^{3/2}$. The initial energy is bounded independently of $r$ and $\ell$. For a sufficiently large fixed $t_0$, $H$ and $E$ are comparable, and

$$|H' + \beta H/t| \le Cr/t + CH/t^2. \tag{5.10}$$

The integrating factors $t^\beta \exp(\pm C/t)$ give

$$ct^{-\beta} - Cr \le H(t) \le Ct^{-\beta} + Cr \qquad (t \ge t_0) \tag{5.11}$$

on this bootstrap interval, with a positive lower constant uniform in $\ell < r$.

Here is the initialization of that constant. On any fixed finite interval the normal coordinate converges in $C^1$ as $r \to 0$, uniformly in $\ell < r$, to the sign-force reference of Lemma 2, and $s \to s_0$. To verify this, isolate its finitely many transverse zeros in disjoint small neighborhoods. Outside $|u| \le 3r$ the normal force is exactly $\operatorname{sgn} u$; in a crossing neighborhood it is bounded uniformly and the reference speed is bounded away from zero. Entry with nearby data therefore crosses the strip of width $6r$ in $O(r)$ time with $O(r)$ velocity change. Between crossings ordinary dependence applies, and the tangential force is $O(r)$. Finite induction proves the assertion. In particular, $E(t_0)$ and $H(t_0)$ tend to strictly positive reference values. The preliminary conditions hold initially, and (5.11) improves them to

$$ct^{-\beta} \le E(t) \le Ct^{-\beta} \qquad (t_0 \le t \le T) \tag{5.12}$$

whenever $rT^\beta$ is small enough. This comparison is not asserted at the singular time zero.

*Occupation of the transition region.* On $|u| \le 3r$, the potential and tangential kinetic energy are $O(r)$. Under (5.12) with $E/r$ large,

$$u'^2 = 2E - A^{-2}z'^2 - 2V \ge ct^{-\beta}. \tag{5.13}$$

Thus there is no turn in this region. A passage beginning near $t_k \ge t_0$ has duration at most $Crt_k^{\beta/2}$: integrate the lower speed bound across width $6r$. The relevant times are comparable since $\beta/2 < 1$ and $rt_k^{\beta/2-1}$ is small.

At its outgoing edge the outward speed is at least $ct_k^{-\beta/2}$. Outside the region the normal force is exactly $\operatorname{sgn} u$, and acceleration is uniformly bounded for $t \ge t_0$. There is consequently a following outward interval of length at least $ct_k^{-\beta/2}$ before that speed halves.

These assigned intervals are disjoint for different passages and satisfy $\int t^\beta \, \mathrm{d}t \geq c t_k^{\beta/2}$. Assign to each complete passage its following such interval when it lies before the observation time $T$. There is at most one last completed passage whose assigned interval is cut off by $T$, as well as a possible last incomplete passage. Each unpaired contribution is bounded by $CrT^{\beta/2}$ and is absorbed by the following bound. The finitely many initial pieces before $t_0$ contribute $O(r)$ by the transverse reference argument. Summing the disjoint assignments gives

$$\big|\{t \in [0,T] : |u(t)| \leq 3r\}\big| \leq Cr(1+T^{\beta+1}), \tag{5.14}$$

$$Q(T) \leq Cr^{3/2}(1+T^{\beta+1}). \tag{5.15}$$

In particular, no completed crossing is charged to a comparison interval beyond the chosen horizon.

At $T = T_r$ the two errors are

$$rT_r^\beta = r^{1-q} \to 0, \qquad Q(T_r) \leq Cr^{3/2-q(1+1/\beta)} \to 0.$$

The upper bound in (5.2) is less than one and gives both limits. The interval for $q$ has width $2\alpha(3-\alpha)/[3(2\alpha+3)] > 0$ throughout the full stated range. For example its midpoint $q = \alpha(\alpha+6)/[3(2\alpha+3)]$ gives

$$Q(T_r) = O(r^{(3-\alpha)/6}), \qquad \omega T_r = A\sqrt{2\kappa}\, r^{-(3-\alpha)/(2(2\alpha+3))}.$$

A first-exit argument closes the three bootstrap conditions $Q \leq 1$, construction-strip retention, and $E \geq C_0 r$: after initialization, the displayed estimates improve all three strictly for sufficiently small $r$ up to $T_r$.

*Tangential decay and the wall guard.* In clock $\tau$, the missing-well coefficient $d(\tau) = 1 - \psi(u/r)$ has integral $Q(T_r) = o(1)$. Lemma 3 gives $|z| + |z_\tau| \leq C|z(0)|\tau^{-\alpha/2}$ for $\tau \geq 1$. The lower inequality in (5.2) is $q > \beta/2$, so $\omega T_r \to \infty$. This proves (5.3) and the bound (5.4). The position tends to its target throughout $1 < \alpha < 3$; the weighted-speed bound tends to zero only for $\alpha > 2$. Also (5.12) and $|u| \leq E + Cr$ give (5.5).

It remains to show that this unconstrained prefix respects the walls. By (2.16) it is uniformly $O(Q(T_r))$ close to

$$s_c(\tau) = m + (s_0 - m)c(\tau). \tag{5.16}$$

The constants do not grow with $\omega T_r$. Since $-\rho_\alpha \leq c \leq 1$, condition (5.1) puts every opposite overshoot of (5.16) strictly inside $[-S, S]$. If $s_0$ is interior, this reference path has a strict uniform wall margin. If $s_0$ is a boundary, choose a small fixed $\tau_0 > 0$ on which $c > 0$. The relative estimate for $z/z(0)$ (when $z(0) \neq 0$) keeps its initial sign up to $\tau_0$. In physical time $(t^\alpha s')' = -\omega^2 t^\alpha \psi(u/r)(s-m)$ then makes motion inward from the boundary; the initial outer plateau has positive coefficient, so it moves strictly inward for

positive small time. After $\tau_0$ the reference path has a strict uniform margin, since $c(\tau) < 1$ for $\tau > 0$ and $c \to 0$. The uniform comparison retains that margin. The case $s_0 = m$ is tangentially constant. Thus there is no positive-time wall contact on the selected prefix. It is the elastic prefix by uniqueness. All estimates were uniform in $\ell < r$, proving the lemma. ∎

# 6 Selection of a nonzero normal endpoint

We first record scalar excursion estimates whose constants do not depend on when an excursion occurs.

**Lemma 9** (Free loss and excursion gain)**.** *A complete free crossing of $[-\ell, \ell]$ loses exactly $2a\ell$ in weighted speed $t|u'|$. If its incoming weighted speed is at most $2a\ell$, it remains in the core forever, with free endpoint in the closed core. For an exterior excursion $x \geq 0$ from zero back to zero in a convex potential $U \geq 0$ with $U(0) = 0$, let $P_{\mathrm{in}}$ and $P_{\mathrm{out}}$ be its outgoing and returning weighted speeds, and $B = \max x$. Then*

$$P_{\mathrm{out}} - P_{\mathrm{in}} \leq 2(3-\alpha)B. \tag{6.1}$$

*For $U(x) = c_* x$ every excursion returns in finite time, and*

$$0 \leq P_{\mathrm{out}} - P_{\mathrm{in}} \leq 2(3-\alpha)B, \quad B \leq P_{\mathrm{in}}/a, \quad P_{\mathrm{out}} \leq K_\alpha P_{\mathrm{in}}, \quad K_\alpha = 1 + \frac{2(3-\alpha)}{a}. \tag{6.2}$$

**Proof.** On a free leg $tu' + au$ is constant. Entering at $+\ell$ with weighted speed $P$, its endpoint is $\ell - P/a$. It crosses the opposite boundary precisely when $P > 2a\ell$, and at that boundary its weighted speed is $P - 2a\ell$. Equality yields asymptotic approach to the opposite boundary, already a final free tail.

Apply (2.8) to an exterior excursion. Convexity implies $xU' - U \geq 0$ and $U \leq xU' \leq BU'$. Integrating the equation multiplied by $t$ gives $\int tU' \,\mathrm{d}t = P_{\mathrm{in}} + P_{\mathrm{out}}$. Therefore

$$\frac{P_{\mathrm{out}}^2 - P_{\mathrm{in}}^2}{2} \leq (3-\alpha)B(P_{\mathrm{in}} + P_{\mathrm{out}}),$$

which proves (6.1). For $U = c_* x$ the moment derivative is $(3-\alpha)c_* tx \geq 0$, giving the other sign. Its force is inward, so the outgoing free comparator bounds $B$ by $P_{\mathrm{in}}/a$. The last inequality follows. Finally the constant-force formula $x' = Ct^{-\alpha} - c_* t/(\alpha+1)$ forces a turn and a finite return. ∎

**Lemma 10** (Arbitrarily small good radii)**.** *For each fixed $r, A, s_0, m$ above, every $\ell \in (0, r)$ gives an elastic block trajectory with a final exact normal free tail*

$$u(t) = q(\ell) + C(\ell)t^{-a}, \qquad q(\ell) \in [-\ell, \ell]. \tag{6.3}$$

*Arbitrarily small such radii satisfy*

$$0 < |q(\ell)| < \ell. \tag{6.4}$$

**Proof.** By Lemma 7 and the barrier (4.10), the normal equation eventually becomes exactly

$$u'' + \alpha u'/t + c_* \operatorname{sgn}(u)\mathbf{1}_{\{|u|>\ell\}} = 0.$$

This is its actual late state, without a reset. Every exterior constant-force excursion returns. Its height tends to zero because normal energy tends to zero. If there were infinitely many complete core crossings, each would lose $2a\ell$ in weighted speed, whereas the following exterior gain is at most $2(3-\alpha)B \to 0$. Eventually the net loss per cycle would be at least $a\ell$, which is impossible for nonnegative weighted speeds. A free inward core leg either parks or crosses the whole core, since its velocity has a fixed sign. It cannot return to its starting wall without such a crossing. These facts exclude additional same-side excursions. Initial incomplete pieces are finite. Thus (6.3) holds and the total number of normal zeros, denoted $N(\ell)$, is finite.

At $\ell = 0$, permanent decoupling instead gives a nontrivial sign-force oscillator. It cannot reach rest at zero at a finite time by Lemma 6. An eventual fixed sign is excluded by integration of its constant-sign force, as in Lemma 2. It has infinitely many transverse zeros. Finite-prefix dependence implies that every prescribed finite number of these zeros persists for all sufficiently small $\ell > 0$. Hence $N(\ell) \to \infty$ in the sense that it eventually exceeds every fixed integer.

Suppose for contradiction that all radii in some $(0, \ell_*)$ have exceptional endpoints $q(\ell) \in \{-\ell, 0, \ell\}$. We show that $N$ is locally constant on this interval. First suppose $q(\ell_0) = \ell_0$. Choose a fixed time $T$ after all zeros and permanent decoupling, far enough along the final free tail that

$$\ell_0/2 < u(T) < \ell_0, \qquad u'(T) > 0.$$

A constant normal boundary tail or finite arrival at rest was excluded in Lemma 6, so these strict choices are possible. Choose $T$ away from tangential nonzero impacts and retain a strict decoupling-energy margin. For nearby radii the state at $T$ is strictly inside its core, with positive position and velocity, and its free comparator $Q_\ell = u_\ell(T) + Tu'_\ell(T)/a$ tends to $\ell_0$.

If $Q_\ell \le \ell$, there is no later zero. If $Q_\ell > \ell$, the outgoing weighted speed of its possible extra boundary bounce is $a(Q_\ell - \ell)$. By (6.2) its next free endpoint is at least

$$\ell - K_\alpha(Q_\ell - \ell) > \ell/2$$

for nearby radii. It then stays positive forever. The retained energy margin keeps all these events in the exact scalar region. This argument permits the extra bounce time to tend to infinity. There are no new zeros after $T$, and the exact earlier count is stable by Lemma 6. Thus $N$ is locally constant at this radius. Reflection handles $q(\ell_0) = -\ell_0$.

If $q(\ell_0) = 0$, choose a fixed late free-core state of nonzero sign. Nearby positions and their comparators lie strictly inside the corresponding cores, and the comparators are

close to zero. These are actual free endpoints. Under the exceptional-only hypothesis they must equal zero, since the two boundary values are excluded by the strict margin. The final tails then keep their signs and add no finite zero. Earlier zeros persist, so $N$ is locally constant here as well. This use of the exceptional-only hypothesis is essential; no unconditional continuity of the total zero count at a zero endpoint is claimed.

An integer-valued locally constant function on the connected interval $(0, \ell_*)$ is constant, contradicting the divergence of $N(\ell)$ at zero. This proves (6.4) for arbitrarily small radii. ∎

# 7 Symmetric smooth completion with exact patches

**Proposition 7.1** (A shape chosen before its height)**.** Fix $A > 0$, a cold tangential start $s_0 \in [-S, S]$, an admissible target $m$, a prescribed positive observation tolerance, and a prescribed positive allowance $\zeta$ for normal overshoot beyond one. There exists a globally nonnegative convex $B \in C^\infty(\mathbb{R}^2)$, with bounded derivatives of every positive order, and numbers

$$0 < R < 1/8, \qquad U_* < 1/2, \qquad k_0 > 0, \qquad 0 < T_o < T_c,$$

with the following properties. On the full guard $I$ it vanishes identically for $|u| \le R$, and is independent of $s$ on a larger normal neighborhood. On the entire outer half-strips over $I$,

$$B_u(s,u) = \operatorname{sgn} u, \qquad B_s(s,u) = k_0(s-m) \quad (|u| > U_*). \tag{7.1}$$

It is even in $u$ and in $s - m$, and satisfies $(s-m)B_s(s,u) \ge 0$ globally. Its elastic cold trajectory from $(s_0, 1)$, with masses $(A^{-2}, 1)$, satisfies

$$|s(T_o) - m| < \text{the prescribed tolerance}, \qquad |u(t)| < 1 + \zeta \quad (t \ge 0), \tag{7.2}$$

and at $T_c$ both

$$U = u(T_c), \qquad Q = u(T_c) + T_c u'(T_c)/a \tag{7.3}$$

have the same nonzero sign and lie strictly in $(-R, R)$. Thus its normal coordinate is exactly free for all $t \ge T_c$ and tends to $Q \neq 0$.

All these choices precede any physical height choice. For a fixed admissible smooth wall $W$, the same finite observation and strict normal certificate persist for the normalized potential $B + \Lambda W$ for every sufficiently large $\Lambda$. Its exact normal free tail then persists for all future time.

**Proof.** First choose $r$ sufficiently small for Lemma 8, for $r < 1/8$, and for the whole-normal allowance. We now describe a fixed enlargement of the constant $C_*$ in (4.8), so this choice is made before smoothing. All local gradients in the construction strip are bounded by one constant $L$ independent of $r, \ell$ and the eventual inner smoothing width. The explicit regularizations below will have uniform value error at most $r$. Therefore

their energy and lower potential bounds will give $|u| \le 1 + \widetilde{C}_* r$ with a fixed $\widetilde{C}_*$. Reserve $\widetilde{C}_* r < \zeta$ now. Choose the early tolerance with a strict margin to allow subsequent finite-prefix perturbation, and put $T_o = T_r$.

Next choose a good radius $\ell < r$ by Lemma 10. Choose a barrier time $T_b > T_o$ away from nonzero tangential impacts, with energy below $c_* r/8$. Farther into the final normal free tail, choose $T_c > T_b$ so that its normal position and comparator have the same nonzero sign and both have absolute values below some $R_* < \ell$. These are strict finite-time data.

*Inner smoothing.* Smooth the sharp inner force monotonically in arbitrarily thin strips of width $\delta$ around $\pm\ell$, keeping it zero on $|u| \le \ell - \delta$ and equal to $c_* \operatorname{sgn} u$ for $|u| \ge \ell + \delta$. The smooth primitive is nonnegative, even, and zero at zero; it changes values by at most $C\delta$ uniformly. All changes occur in the scalar product region, so convexity and both symmetries remain. Lemma 6, rather than a false $C^1$ approximation across a sharp jump, preserves the observation, the barrier, and the late normal data. Fix this smoothing, calling the smooth local potential $V_\delta$.

*Global supporting extension.* Define $H$ to be the supremum of all affine supporting functions of $V_\delta$ based in $D_m$, together with the zero plane. Every slope has norm at most $L$. At a fixed $x_0 \in D_m$ each supporting plane has value at most $V_\delta(x_0)$, so $H(x) \le V_\delta(x_0) + L|x - x_0|$. Hence $H$ is finite, globally convex, nonnegative, and $L$-Lipschitz. Convexity on the strip puts all these planes below $V_\delta$ there, while the plane based at each point gives equality. Thus $H = V_\delta$ throughout the whole unbounded strip. Reflecting any supporting plane in $u = 0$ or $s = m$ produces another, so $H$ has both symmetries.

For $\lambda > 0$ take the Moreau envelope

$$H_\lambda(x) = \inf_p \left( H(p) + \frac{|x - p|^2}{2\lambda} \right).$$

Its unique minimizing point satisfies $(x - p)/\lambda \in \partial H(p)$ and therefore $|x - p| \le \lambda L$. Completing the square in the Lipschitz lower bound gives globally

$$H - \lambda L^2/2 \le H_\lambda \le H. \tag{7.4}$$

Convolve with a nonnegative compactly supported smooth product probability kernel $\rho$, even in each coordinate and centered at zero, and set $B = H_\lambda * \rho$. It is smooth, convex, nonnegative, and has both symmetries. Its gradient is bounded by $L$. For every order $j \ge 1$, write $D^j B$ as a convolution of a bounded first weak derivative of $H_\lambda$ with an order $j - 1$ derivative of the compact kernel. This proves global boundedness at every positive order. Symmetry and one-dimensional convexity of each $s$-slice imply the inward sign globally. On compact subsets of the strip the proximal points stay inside it, so $\nabla H_\lambda(x) = \nabla V_\delta(p(x)) \to \nabla V_\delta(x)$. Compact convolution preserves this local $C^1$ convergence. Lemma 4 now applies to the fixed smooth shape.

*Exact inner patches.* Keep $\lambda L$ and the kernel support smaller than every retained tangential and normal patch margin. In the original inner product region the potential

depends only on $u$. A proximal minimizer there must keep its tangential coordinate equal to the evaluation coordinate, since only the squared distance depends on that coordinate. Thus the envelope is exactly one-dimensional on the smaller product region, and product convolution preserves independence of $s$. The envelope of a nonnegative function has exactly the same zero set: both nonnegative summands in its minimization must vanish for a zero. Convolution is zero whenever all its translates stay in that zero set. Choose

$$R_* < R < \ell$$

inside the eroded smoothed flat core. These local choices are made on the full guard $I$, not just the hard-wall interval.

*Exact complete outer half-strips.* Let $\sigma_s, \sigma_u$ bound the kernel's coordinate support. The distance of $I$ from the construction-strip boundary exceeds $\lambda L + \sigma_s$. For every $|u| > U_* := 3r + \lambda L + \sigma_u$, the entire proximal and convolution calculation stays in the corresponding outer half-strip. On its positive side $V_\delta = u + h_0(s-m)^2 + C_\delta$, $h_0 = \kappa r$, so direct minimization gives

$$H_\lambda(s,u) = u - \lambda/2 + \frac{h_0}{1+2\lambda h_0}(s-m)^2 + C_\delta. \tag{7.5}$$

Centered product convolution changes only the constant. Reflection gives the negative side. Thus (7.1) holds with $k_0 = 2h_0/(1+2\lambda h_0) > 0$ on the entire stated half-strips. Choose the scales so $U_* < 1/2$. In particular, nonnegativity and the outward unit normal slope imply

$$B(s,u) \geq |u| - U_* \quad (s \in I,\ |u| \geq U_*). \tag{7.6}$$

*Energy budgets and the late certificate.* If $\sigma$ bounds the Euclidean kernel support, then $|B - H| \leq \lambda L^2/2 + L\sigma$ globally. Together with the inner smoothing error, choose this at most $r$. On the whole guard we obtain $B \geq |u| - Cr$, and the cold energy is at most $1 + Cr$. This proves the reserved all-time normal bound with the fixed $\widetilde{C}_*$, independently of finite-prefix approximation. Choose the value errors also small enough that on $u = \pm b_0$ the potential exceeds $3c_*r/8$, while the completed orbit's energy at $T_b$ is below $c_*r/4$. Keep the whole set $I \times [-b_0, b_0]$ inside the exact product patch after all proximal and convolution displacements. Energy then prevents later normal exit from that patch. The choice of $T_b$ away from impacts permits ordinary kinetic convergence there; uniform mechanical-energy convergence would also suffice.

Finally finite-prefix dependence preserves both strict normal inequalities at $T_c$ and the observation at $T_o$. Only normal position and velocity are needed at $T_c$, which may be a tangential impact. The free continuation

$$u(t) = Q + (U - Q)(T_c/t)^a$$

stays between the certified same-sign values $U, Q$ in $(-R, R)$. The hard tangential coordinate stays in $I$. This is therefore an exact solution of the coupled equation for all future time, and uniqueness identifies it with the actual continuation.

For the last assertion apply Lemma 4 to the fixed $B$ and let $\Lambda \to \infty$. Cold normalized energy is $B(s_0, 1)$, independent of $\Lambda$. The wall values at the endpoints of $I$ then confine the soft coordinate to $I$ for all time when $\Lambda$ is large. The finite observation and normal certificate persist. The exact free normal continuation stays in the zero core, while the scalar wall continuation stays in $I$ by energy. Together they solve the full soft equation forever. All smoothing scales and derivative norms have remained fixed. ∎

# 8 Scalar wall tails and every boundary release case

Fix once and for all an even convex $W \in C^\infty(\mathbb{R})$, zero exactly on $[-S, S]$, coercive, with bounded derivatives of every positive order, and with

$$W(S + \eta) > 2. \tag{8.1}$$

Such a wall is obtained by prescribing an even nonnegative smooth second derivative, positive on $S < |s| < S + \eta/2$, flat at those endpoints, and zero elsewhere; integrate from the flat interval and scale. Outside the transition its tails are affine with positive outward slope.

**Lemma 11** (Exhaustive scalar wall tails)**.** *Every solution of $s'' + \alpha s'/t + W'(s) = 0$ prescribed at a positive time is, after a finite transient, of one of these forms:*

1. *an exact free tail $s(t) = s_\infty + Ct^{-a}$ in the flat interval, with $s_\infty \in [-S, S]$, including constant endpoints;*

2. *a monotone exterior approach to $S$ or to $-S$.*

*On a right exterior tail $x = s - S > 0$, the following quantities tend to zero:*

$$x(t), \qquad t|x'(t)|, \qquad t^2 W'(S + x(t)), \qquad t^2 W(S + x(t)). \tag{8.2}$$

**Proof.** Energy and coercivity bound position and speed. By Lemma 7 the energy tends to zero, so exterior heights tend to zero. Each complete free crossing of $[-S, S]$ loses $2aS$ in weighted speed. Each completed exterior excursion gains at most $2(3 - \alpha)B$ by Lemma 9. Thus there can be only finitely many complete crossings: otherwise a fixed positive net loss per cycle would eventually exhaust the nonnegative weighted speed.

To check that these counts give an exhaustive classification, an exterior outward leg turns in finite time. Otherwise it remains at a positive exterior distance with a positive lower bound on $|W'|$, and the weighted momentum equation forces a turn. After the turn, the inward weighted momentum cannot turn outward again before entry. If this inward leg never enters, energy decay forces its position to approach that boundary monotonically. If it enters, its free velocity has a fixed sign; it either stays in the core forever or

crosses the entire core and begins an excursion at the other side. It cannot produce repeated tiny same-side excursions without complete crossings. These facts prove the stated alternatives. A finite arrival at a smooth boundary at zero velocity from a nonconstant motion is excluded by ordinary positive-time uniqueness. A trajectory already constant there is explicitly included.

On a right exterior asymptote put $p = -t^\alpha x' > 0$. It is nondecreasing and

$$x(t) = \int_t^\infty p(v) v^{-\alpha} \, \mathrm{d}v \geq p(t) t^{1-\alpha}/a.$$

Thus $t|x'| \leq ax \to 0$. For late $t$, convexity and monotonicity on the tail give

$$p(t) \geq \int_{t/2}^t v^\alpha W'(S + x(v)) \, \mathrm{d}v \geq c_\alpha t^{\alpha+1} W'(S + x(t)).$$

It follows that $t^2 W'(S + x(t)) \leq Cx(t) \to 0$. Finally $W(S + x) \leq xW'(S + x)$, proving the last limit. ∎

We next give the estimates that make a small new well release these actual tails. Let $k > 0$ and $m \in (-S, S)$ be fixed, and add the tangential force $\varepsilon k(s - m)$. The natural time scale is a fixed multiple of $\varepsilon^{-1/2}$. All limits below are unchanged by such a fixed multiplier. At each sufficiently late fixed physical time $T$, the perturbed data are assumed to converge to the actual old tail data as $\varepsilon \to 0$. We never replace these perturbed data by the limiting ones.

**Lemma 12** (Early release and vanishing weighted kick)**.** *For the scalar wall tails in Lemma 11, a sufficiently small added well has the following behavior in its initial one-sided phase. From a right exterior asymptote it enters the flat interval at time $t_e$ with*

$$t_e \sqrt{\varepsilon} \longrightarrow 0, \qquad P_e = -t_e s'_\varepsilon(t_e) \longrightarrow 0. \tag{8.3}$$

*A free tail approaching $S$ from inside either remains on the flat side during that phase or makes one early outward excursion and returns with the same properties. A stationary tail at $S$ has the same conclusion, allowing initial perturbations on either side and either velocity sign. The reflected assertions hold at $-S$. For a strict interior free endpoint no wall release is necessary. After any release, on a sufficiently small fixed slow interval there is no second wall excursion, and the actual position and slow velocity at each fixed positive slow time converge to those of the regular cold well equation started at the old tangential limit.*

**Proof.** Choose a small right boundary strip lying entirely above $m$. On it the new inward force is bounded above by $C\varepsilon$ and below by $c_0\varepsilon$ for fixed positive constants. We verify every type of tail and then the last convergence assertion.

*Exterior asymptote.* Write $x = s - S$ and $d = S - m > 0$. On the exterior the potential after subtracting its boundary value is

$$U_\varepsilon(x) = W(S + x) + \varepsilon k(dx + x^2/2), \qquad U'_\varepsilon \geq \varepsilon k d.$$

At a fixed late $T$ the actual perturbed data have $x(T) > 0$, $x'(T) < 0$ for small $\varepsilon$. The weighted velocity decreases until entry, so $0 \le x(t) \le x(T)$. With

$$M(t) = \tfrac{1}{2}(tx' + ax)^2 + t^2 U_\varepsilon(x),$$

convexity and (2.8) give $M' \le (3-\alpha)tU_\varepsilon \le (3-\alpha)x(T)tU'_\varepsilon$. At first entry $t_e$, integration of the equation times $t$ gives

$$\int_T^{t_e} tU'_\varepsilon \,\mathrm{d}t = Tx'(T) + P_e + ax(T) \le P_e + ax(T).$$

Hence

$$\frac{P_e^2}{2} \le M(T) + (3-\alpha)x(T)[P_e + ax(T)], \qquad P_e \le C_\alpha\big(x(T) + \sqrt{M(T)}\big). \tag{8.4}$$

For fixed $T$ the initial moment converges to the old moment; by (8.2) this old moment tends to zero as $T \to \infty$. Given any tolerance, choose a sufficiently late fixed $T$ and then a sufficiently small height. This proves the ordinary limit $P_e \to 0$ for the actual family.

The positive lower new force gives the Volterra bound, until entry,

$$x(t) \le x(T) - \frac{\varepsilon k d}{a} \int_T^t v[1 - (v/t)^a] \,\mathrm{d}v. \tag{8.5}$$

For $t = \delta/\sqrt{\varepsilon}$ the subtracted term tends to $kd\delta^2/[2(\alpha+1)]$. For any fixed $\delta > 0$, first choose $T$ so $x(T)$ is smaller than half this quantity and then choose $\varepsilon$ small. This forces entry before that time, proving both finiteness of entry and the first limit in (8.3). The two choices prove a limit of one actual family; no new data are assigned at $T$.

*A boundary approached from inside.* For the old free tail, late data have $x(T) < 0$, $x'(T) > 0$, and $x(T) + Tx'(T)/a = 0$. The corresponding perturbed comparator $Q_\varepsilon = x(T) + Tx'(T)/a$ tends to zero. In the initial strip (2.7) makes the comparator nonincreasing. If an outward exit occurs at $t_x$, it occurs before the weighted velocity turns inward. At the exit the outgoing weighted speed is at most $a(Q_\varepsilon)_+$ and its prospective free height is at most $(Q_\varepsilon)_+$.

The timing also must be checked. Until such an exit the initial velocity is positive, and the free part of the positive-time Volterra formula is at most $Q_\varepsilon$. Its lower-force term gives

$$0 \le Q_\varepsilon - \frac{\varepsilon c_0}{a} \int_T^{t_x} v[1 - (v/t_x)^a] \,\mathrm{d}v. \tag{8.6}$$

If $t_x \ge 2T$, the integral is at least $ct_x^2$, so $t_x\sqrt{\varepsilon} \le C\sqrt{(Q_\varepsilon)_+} \to 0$. Exits before $2T$ also occur at vanishing slow time. On the subsequent exterior excursion the height is at most $(Q_\varepsilon)_+$, and (6.1) makes the returning weighted speed at most $a(Q_\varepsilon)_+ + 2(3-\alpha)(Q_\varepsilon)_+$, which tends to zero. Starting the lower-force formula at this already early exit bounds the future free displacement by $(Q_\varepsilon)_+$ and forces return before every fixed positive slow time.

Thus return is early as well. If the velocity turns on the flat side instead, the ordinary bounded-force formula applies there.

These arguments were made only in the initial one-sided strip. They close there: choose the old tail's displacement and weighted speed small by taking $T$ late, and then choose a small fixed slow cutoff. The bounded-force displacement estimate prevents reaching the inward edge of the strip before that cutoff. After an exterior return use its vanishing weighted speed and the same estimate. This proves the required initial control without asserting that later well oscillations can never hit a wall.

*A stationary boundary tail.* Now the actual perturbed data at a fixed $T$ satisfy $x(T), x'(T) \to 0$, with either sign. For exterior data define

$$B_\varepsilon = x(T) + \frac{T(x'(T))_+}{a} \longrightarrow 0.$$

The inward force bounds the exterior height up to entry by $B_\varepsilon$, even if there is first an outward turn. The same moment calculation, now retaining the initial velocity term, gives

$$\frac{P_e^2}{2} \le M(T) + (3-\alpha)B_\varepsilon[P_e + Tx'(T) + ax(T)] \le M(T) + (3-\alpha)B_\varepsilon[P_e + aB_\varepsilon]. \quad (8.7)$$

Here $M(T) \to 0$ for fixed $T$, so $P_e \to 0$. The lower-force formula with free upper position $B_\varepsilon$ proves early entry. For flat-side data with outward velocity the preceding inside-start proof applies; with nonpositive velocity, use the bounded-force formula directly. Reflection supplies all three left-boundary cases.

*The exact no-reset bridge.* Let the slow scale be $L = \sqrt{A/\varepsilon}$ for a fixed $A > 0$, write $\tau = t/L$, and allow $k = Ak_0$, as required below. After right-wall release put $\tau_e = t_e/L$. On the initial flat-side interval the actual trajectory obeys

$$\begin{aligned} s_\varepsilon(\tau) = S &- \frac{P_e}{a}[1 - (\tau_e/\tau)^a] \\ &- \frac{A^2 k_0}{a} \int_{\tau_e}^{\tau} v[1 - (v/\tau)^a](s_\varepsilon(v) - m)\,\mathrm{d}v. \end{aligned} \quad (8.8)$$

Before reaching $m$, its signed weighted velocity points inward, and this formula bounds its distance from the wall by $P_e/a + C\tau^2$. A small fixed cutoff therefore keeps it in the one-sided strip and prevents a second excursion there. The homogeneous displacement tends to zero, and the missing integral on $[0, \tau_e]$ also tends to zero. Volterra dependence with the fixed coefficient $A^2k_0$ proves convergence to the regular cold well solution from $S$. At a fixed positive $\delta$, differentiating the actual formula gives

$$s'_\varepsilon(\delta) = -P_e\tau_e^{\alpha-1}\delta^{-\alpha} - A^2k_0\delta^{-\alpha}\int_{\tau_e}^{\delta} v^\alpha(s_\varepsilon(v) - m)\,\mathrm{d}v. \quad (8.9)$$

The first term tends to zero and the second to the cold derivative. The slow entry velocity $-P_e/\tau_e$ itself need not be bounded; it has never been replaced by zero.

For a strict interior free endpoint, take $T$ late with a strict wall margin. Its old free comparator is $s_\infty$, and the perturbed comparator converges to it. The positive-time free formula plus a bounded $O(\varepsilon)$ force controls the growing interval up to $\delta L$, excluding wall contact for small $\delta$. The free homogeneous part and its slow derivative tend to zero at each positive slow time, since $a > 0$. The same Volterra argument identifies the cold limit from $s_\infty$. In the inside-start cases with no excursion, it gives the same conclusion from the original data at $T$: first make the old tail errors small by choosing $T$, then decrease $\varepsilon$. Thus all the classified cases lead to the asserted convergence. This proves the lemma. ∎

# 9 Insertion into the original cold trajectory

Let $G = (-S - \eta, S + \eta)$ be the interior of the fixed guard $I$. In the induction, every old block will be nonnegative and zero on $G$ times a normal interval. We keep a common energy bound, so all actual physical trajectories stay in a compact subset of $G$.

**Proposition 9.1** (Actual-tail insertion)**.** Suppose

$$F_{\text{old}}(s, y) = W(s) + \sum_{j<n} f_j(s, y)$$

is a finite sum of globally smooth nonnegative convex functions with bounded derivatives of positive order. Suppose its original cold solution from $(0, 1)$ has an eventual exact normal free tail

$$y_{\text{old}}(t) = q_{\text{old}} + Ct^{-a}, \qquad 0 < A := |q_{\text{old}}| < R_{\text{old}}, \tag{9.1}$$

and all old blocks vanish on $G \times [-R_{\text{old}}, R_{\text{old}}]$. Assume the original and sufficiently small perturbed trajectories are confined tangentially to a common compact subset of $G$ by an initial-energy bound strictly below the wall barrier. Let $s_\infty \in [-S, S]$ be the tangential limit supplied by Lemma 11. Fix an admissible target $m$ and $\delta > 0$.

There is one fixed smooth shape $B$ as in Proposition 7.1 such that, for every sufficiently small $\varepsilon > 0$, addition of

$$f_\varepsilon(s, y) = \varepsilon AB(s, y/A) \tag{9.2}$$

has both the following effects on the original cold trajectory:

1. at a time $t_o = \sqrt{A/\varepsilon}\, T_o$ its tangential coordinate satisfies $|s(t_o) - m| < 2\delta$;

2. it has a later exact normal free tail with endpoint $q_{\text{new}}$ and common zero radius $R_{\text{new}}$ satisfying

$$0 < |q_{\text{new}}| < R_{\text{new}} < A/8. \tag{9.3}$$

   Its eventual tangential equation is exactly the scalar $W$ equation.

Every finite collection of upper bounds on positive-order derivatives of $f_\varepsilon$ can be met by further decreasing $\varepsilon$, with this same shape.

**Proof.** Reflect the old normal coordinate if necessary. Since $B$ is even in its normal variable, we can write $u = \operatorname{sgn}(q_{\mathrm{old}})y/A$ and use positive normalized starting amplitude one. Choose the block of Proposition 7.1 from $(s_\infty, 1)$, with observation error less than $\delta$ and a whole-normal allowance $\zeta$ so small that

$$A(1+\zeta) < R_{\mathrm{old}} \tag{9.4}$$

with a strict margin. Its shape, core radius $R < 1/8$, exact outer threshold $U_* < 1/2$, derivative norms, and times $T_o, T_c$ are now fixed.

Put $L = \sqrt{A/\varepsilon}$ and $\tau = t/L$. Inside the old normal product core the normalized equations are exactly

$$u'' + \alpha u'/\tau + B_u(s,u) = 0, \tag{9.5}$$

$$s'' + \alpha s'/\tau + A^2 B_s(s,u) + (A/\varepsilon)W'(s) = 0. \tag{9.6}$$

Primes here denote slow derivatives. The masses are $(A^{-2}, 1)$ and the normalized potential is

$$B(s,u) + \frac{W(s)}{\varepsilon A}. \tag{9.7}$$

Thus the penalty parameter in Lemma 4 is $\varepsilon A$; the mass-divided tangential coefficient is $A/\varepsilon$. These are different, consistent normalizations.

*The exact early normal leg.* At every fixed late physical time $T$, the actual perturbed cold data converge to the old data by Lemma 1, since the added force is uniformly $O(\varepsilon)$ with fixed shape. Choose $T$ late enough that the reflected old tail lies in a compact subinterval of

$$(A/2, \min(3A/2, R_{\mathrm{old}})).$$

As long as the perturbed trajectory stays in this interval, the complete outer half-strip formula gives the exact physical normal force $\varepsilon$ and tangential well $\varepsilon A k_0(s-m)$. Its reflected normal coordinate $z$ is therefore exactly

$$z_\varepsilon(t) = Q_\varepsilon + C_\varepsilon t^{-a} - \frac{\varepsilon t^2}{2(\alpha+1)}, \qquad Q_\varepsilon \to A, \tag{9.8}$$

where $C_\varepsilon$ tends to the reflected old tail coefficient. For a small fixed slow cutoff $\delta_0 > 0$, first take $T$ late and then $\varepsilon$ small. Formula (9.8) stays in the chosen compact interval for $T \le t \le \delta_0 L$. Together with the assumed global tangential guard, this is a first-exit proof that the plateau and the old zero core remain valid throughout that growing interval. In normalized coordinates its position and slow derivative at any fixed positive slow time converge to those of

$$u_c(\tau) = 1 - \frac{\tau^2}{2(\alpha+1)}. \tag{9.9}$$

The old homogeneous term disappears because $L^{-a} \to 0$; no physical or slow velocity has been reset.

*Tangential release and a bounded-energy starting state.* On the same exact plateau the old blocks exert no force, and the actual tangential equation is precisely the scalar wall plus $\varepsilon A k_0(s-m)$ equation. Apply Lemma 12 to every case in Lemma 11: a strict interior free limit, a boundary approached freely from inside, a stationary boundary, or an exterior asymptote. Release, when needed, occurs at $\tau_e \to 0$ with weighted kick $P_e \to 0$. The exact formulas (8.8)–(8.9) identify both normalized position and velocity at a fixed small $\delta_0 > 0$ with the cold plateau solution

$$s_c(\tau) = m + (s_\infty - m)c_\alpha(A\sqrt{k_0}\,\tau). \tag{9.10}$$

Choose $\delta_0 < \min(T_o, T_c)$ small enough for these plateau arguments. The limiting state is strictly inside $(-S, S)$ at $\delta_0$: for an interior start this follows by continuity; for a boundary start its nonzero initial inward force makes it strictly interior at all sufficiently small positive times. For all sufficiently small heights, the actual state at this fixed slow time therefore has $W = 0$ exactly. Both normalized velocities converge, and all old blocks vanish. Its normalized total energy in (9.7) is bounded and converges to the cold hard-wall energy at $\delta_0$.

This is the bounded-energy bridge needed for a penalty limit. The quantity

$$W(s_\varepsilon(T))/(\varepsilon A)$$

at a fixed old physical time may be unbounded. No assumption that it is bounded, or that a large initial wall energy can be discarded, has entered the proof.

*Continuation through all later wall events.* From the actual state at slow time $\delta_0$, consider the surrogate normalized problem containing just $B$ and the penalty $W/(\varepsilon A)$. The positive-time interior-data version of Lemma 4, on an auxiliary horizon beyond $T_c$, identifies its limit with the continuation of (9.9)–(9.10), hence with the selected elastic cold orbit. Positions and normal velocities converge uniformly; tangential velocities converge away from nonzero impacts. Its bounded normalized energy gives vanishing wall penetration. The hard normal path has the fixed old-core margin (9.4), and the hard tangential path has positive distance from the boundary of $G$. The surrogate therefore stays inside the old product core through $T_c$ for small heights.

Until a first exit, the actual and surrogate equations have the same data and exactly the same forces. Their uniqueness and the surrogate's strict margins exclude that exit. Thus they agree through $T_c$. This step justifies deletion of old forces during the whole later prefix, including possible repeated wall impacts; retention of the early steering observation alone would not do so. At $T_o$ the position observation is preserved with error less than $2\delta$. At $T_c$, uniform normal convergence preserves both strict same-sign normal inequalities (7.3). The time $T_c$ may be a tangential impact. Neither tangential velocity convergence nor pointwise vanishing of wall energy is needed there.

*An infinite exact physical tail.* At physical time $t_c = LT_c$ the normal coordinate and comparator are in $(-AR, AR)$, with the same nonzero sign. Define the signed physical

comparator $q_{\mathrm{new}}$ accordingly. Then $0 < |q_{\mathrm{new}}| < AR < A/8$; put $R_{\mathrm{new}} = AR$. All old blocks and the new block vanish at this state. Its physical energy is at most $\varepsilon A$ times a bounded normalized energy, including any wall energy. Starting from its actual tangential state, the scalar $W$ continuation therefore remains in $G$ for all future time when the height is small. Pair it with

$$y(t) = q_{\mathrm{new}} + [y(t_c) - q_{\mathrm{new}}](t_c/t)^a. \tag{9.11}$$

This normal path stays between its two certified endpoint values, hence in $[-AR, AR]$. On the product with $G$, every block has zero gradient. The paired continuation solves the full original smooth equation, and uniqueness identifies it with the actual orbit forever. Lemma 11 supplies the next tangential tail.

Finally, for the fixed $A$ and fixed shape, each number

$$C_j = \|D^j[AB(s, y/A)]\|_\infty \qquad (j \geq 1)$$

is finite. The corresponding derivative norm of $f_\varepsilon$ is $\varepsilon C_j$. Every finite list of upper bounds can therefore be imposed after all shape choices. The observation time $LT_o$ tends to infinity as the height decreases. All assertions hold for every sufficiently small positive height, not just along a selected sequence. ∎

# 10 One static objective and its nonconvergent cold orbit

We now prove Theorem 1. Fix $d > 0$ sufficiently small that both targets $m_n = (-1)^n d$ satisfy (5.1), and put

$$\delta_n = \min(d/16, 2^{-n-4}). \tag{10.1}$$

Keep the physical wall $W$ fixed throughout. We construct nonnegative convex smooth blocks $f_n$ with a common tangential guard and nested exact normal zero cores.

Start with $F_0 = W$, whose cold solution from $(0, 1)$ is constant. It has $A_0 = 1$ and any chosen normal zero radius $R_0 > 1$; there are no old blocks. Suppose $F_{n-1} = W + \sum_{j<n} f_j$ has a nonzero exact normal free endpoint $q_{n-1}$ strictly inside its common zero radius $R_{n-1}$. Write $A_{n-1} = |q_{n-1}|$. Its eventual tangential equation is the scalar wall equation. Select a normalized shape by Proposition 9.1 for target $m_n$ and tolerance $\delta_n$, then choose its height $\varepsilon_n$.

At stage $n$, require

$$\|D^j f_n\|_\infty \leq 2^{-n-2} \qquad (1 \leq j \leq n+2). \tag{10.2}$$

These requirements ensure the common energy guard needed in the insertion proposition. Indeed $f_n(0, 0) = 0$, so the line integral of its gradient gives $f_n(0, 1) \leq 2^{-n-2}$.

Consequently every partial sum, and later the final sum, has initial cold energy at most

$$\sum_{n=1}^{\infty} 2^{-n-2} = 1/4. \tag{10.3}$$

Since every block is nonnegative, the tangential coordinate stays in the compact interval

$$K_s = \{s : W(s) \le 1/4\} \Subset G \tag{10.4}$$

by (8.1). This also holds for all sufficiently small candidate heights at each stage. Thus the insertion hypothesis is satisfied without assuming a wall confines arbitrary-energy trajectories.

Choose the height small enough for all insertion conclusions and so that its observation time $t_n$ exceeds $n$ and the previous retained certificate time by at least one. Retain

$$|s_n(t_n) - m_n| < 2\delta_n, \tag{10.5}$$

where $s_n$ is the tangential coordinate of the exact cold $F_n$ trajectory. Retain its new exact normal free tail and core. The insertion gives

$$0 < A_n < R_n < A_{n-1}/8 < R_{n-1}. \tag{10.6}$$

Hence all previous blocks vanish in the new common product core. The construction is ready for the next stage.

We impose additional force budgets to protect every observation. By (10.2), all partial sums have gradient Lipschitz constant at most

$$L_0 = \|W''\|_\infty + 1/4. \tag{10.7}$$

After stage $n$, choose $\eta_n > 0$ so small that $B_{t_n}(\eta_n) < \delta_n$ in (2.4). For every later $k > n$ require

$$\|\nabla f_k\|_\infty \le 2^{-(k-n+1)}\eta_n. \tag{10.8}$$

At each stage there are only finitely many such old tolerances, finitely many derivative orders in (10.2), and finitely many insertion and observation-time requirements. They are all satisfied by a sufficiently small positive height with the selected shape unchanged. In particular, there is no competing lower height bound from a subsequently chosen smoothing width.

Define one static function

$$F(s, y) = W(s) + \sum_{n=1}^{\infty} f_n(s, y). \tag{10.9}$$

The gradient series converges uniformly. Integrating along a segment from the common zero $(0, 0)$ proves local uniform convergence of the value series. For each fixed order $j \ge 1$,

all sufficiently late terms satisfy the summable bound (10.2); the finitely many earlier derivative norms are finite. Therefore every positive-order derivative series converges uniformly. Local differentiation of these uniformly convergent series proves that $F$ is globally smooth with every positive-order derivative globally bounded. Convexity and nonnegativity pass to the local uniform limit, and (10.7) is a global Hessian bound.

The minimizer set is exactly the one in (1.2). All terms vanish on $[-S, S] \times \{0\}$. For $|s| > S$, the wall is strictly positive. For $|s| \le S$ and $y \neq 0$, choose a late block with $|y|/A_{n-1} > U_{*,n}$, possible by (10.6) and $U_{*,n} < 1/2$. Its complete outer half-strip bound (7.6) gives

$$f_n(s, y) \ge \varepsilon_n A_{n-1}(|y|/A_{n-1} - U_{*,n}) > 0.$$

This proves exactness, without any estimate on stage-dependent affine intercepts.

Lemma 1 gives a unique global regular cold solution for $F$. Its initial energy is at most $1/4$, so its tangential coordinate lies in (10.4). On this compact guard the first block has a complete affine normal tail; in particular $f_1(s, y) \ge \varepsilon_1|y| - C_1$ for one fixed finite $C_1$. Since $F \ge W + f_1$, energy bounds $y$ as well. Thus the final orbit is bounded.

For each retained stage $n$, summing (10.8) gives

$$\|\nabla F - \nabla F_n\|_\infty \le \eta_n \sum_{j=1}^{\infty} 2^{-(j+1)} = \eta_n/2.$$

Compare the final and partial-sum cold trajectories directly by Lemma 1. Equations (10.5) and (2.4) yield

$$|s(t_n) - m_n| < 3\delta_n \longrightarrow 0. \tag{10.10}$$

This proves (1.3), since $t_n \to \infty$. There is no accumulated succession of solution errors: for each $n$ the complete future force tail is compared at once with $F_n$, always from the same original cold data.

Finally, apply the ordinary bounded-orbit part of Lemma 7 to $F$. Its mechanical energy tends to zero, hence $F(X(t)) \to 0$ and $X'(t) \to 0$. Boundedness and continuity then imply $\operatorname{dist}(X(t), \operatorname{argmin} F) \to 0$: otherwise a convergent subsequence at positive distance would have a nonminimizing limit of zero potential. In particular $y(t) \to 0$. Thus the observations actually give the two distinct minimizing cluster points $(d, 0)$ and $(-d, 0)$. This completes the proof of Theorem 1.

**Corollary 10.1.** *The trajectory in Theorem 1 has infinite length. Its set of limiting positions contains* $[-d, d] \times \{0\}$.

**Proof.** For large $n$ consecutive observations have tangential separation at least $d$. The length on $[t_n, t_{n+1}]$ is therefore at least $d$, and summation proves infinite length. For any $\xi \in (-d, d)$, continuity supplies times between consecutive oppositely signed observations at which $s = \xi$. These times tend to infinity and $y \to 0$, so $(\xi, 0)$ is a limiting position. The endpoints follow from the selected subsequences. This does not assert that the limiting set equals this subsegment. ∎

# 11 The universal damping threshold

For contrast with the bounded compact-minimizer construction, the lower endpoint admits a simple unbounded example.

**Proposition 11.1** (Cold divergence for low damping)**.** For every $0 < \alpha \le 1$ there is a nonnegative convex $g \in C^\infty(\mathbb{R})$ with every positive-order derivative bounded, $0 \le g'' \le 1$, and $\operatorname{argmin} g = (-\infty, 0]$, whose regular cold solution from $x(0) = 1$, $x'(0) = 0$ tends to $-\infty$.

**Proof.** Choose a nonnegative smooth $\rho$ supported on $[0, 2]$, positive on $(0, 2)$, flat at both endpoints, and bounded by one. Set $g'' = \rho$, $g'(0) = g(0) = 0$. Then $g = 0$ on the negative half-line, $g' > 0$ on the positive half-line, and all asserted regularity and convexity properties follow. Its first derivative is bounded, being an integral of a compactly supported function.

For every $\alpha > 0$, including one, the cold equation has the nonsingular double-integral form

$$x(t) = 1 - \int_0^t v^{-\alpha} \int_0^v w^\alpha g'(x(w)) \, \mathrm{d}w \, \mathrm{d}v.$$

The integral operator has local contraction constant at most $\|g''\|_\infty t^2/[2(\alpha+1)]$; differentiation gives the cold regularity and global continuation as before. No formula containing $1/(\alpha - 1)$ is used at $\alpha = 1$. Its velocity is

$$x'(t) = -t^{-\alpha} \int_0^t v^\alpha g'(x(v)) \, \mathrm{d}v.$$

If $x$ remained positive forever, the integral would, after some fixed short initial interval, be bounded below by a fixed positive constant. Thus $x'(t) \le -Ct^{-\alpha}$, contradicting positivity because this negative upper bound has divergent time integral for $\alpha \le 1$. It reaches zero at a finite time $t_c$ with strictly negative velocity. Thereafter the exact continuation is

$$x(t) = -C_c \int_{t_c}^t v^{-\alpha} \, \mathrm{d}v, \qquad C_c = -t_c^\alpha x'(t_c) > 0.$$

It stays negative, solves the zero-force equation, and tends to $-\infty$. Uniqueness identifies it with the original solution. The nonzero velocity at $t_c$ was generated by the force from the original zero-velocity data. ∎

For completeness we include the known convergence implication, using the energy-difference mechanism of Jang and Ryu [2] at critical damping and the same calculation above it.

**Proposition 11.2** (Convergence at and above critical damping)**.** Let $\alpha \ge 3$ and let $F : \mathbb{R}^k \to \mathbb{R}$ be convex with globally Lipschitz gradient and nonempty minimizer set. For every $x_0 \in \mathbb{R}^k$, the regular cold solution with $X(0) = x_0$ and $X'(0) = 0$ converges to a minimizer.

**Proof.** Subtract $\min F$, so $F \geq 0$, and set $a = \alpha - 1$. For each minimizer $z$ define

$$\mathcal{E}_z(t) = t^2 F(X(t)) + \tfrac{1}{2}|tX'(t) + a(X(t) - z)|^2.$$

Direct differentiation gives

$$\mathcal{E}_z' = t\big((3-\alpha)F - a(\langle X - z, \nabla F(X)\rangle - F)\big) \leq 0.$$

It therefore has a finite limit. In particular $F(X(t)) = O(t^{-2})$ and $w(t) = tX'(t) + a(X(t) - z)$ is bounded. Integrating $(t^a(X - z))' = t^{a-1}w(t)$ from zero shows that $X$ is bounded, since $a > 0$ and $t^a(X(t) - z) \to 0$ at zero. Hence there is a cluster point, and every cluster point is a minimizer.

For any two minimizers $z_1, z_2$ put $h(t) = |X(t) - z_1|^2 - |X(t) - z_2|^2$. Cancellation of the common potential and kinetic terms gives

$$\mathcal{E}_{z_1} - \mathcal{E}_{z_2} = \frac{a}{2}\big(th' + ah\big).$$

The left side converges. If $th' + ah \to \ell$, the integrating factor $t^a$ gives $h(t) \to \ell/a$: split its defining integral at a fixed late time, bound the finite prefix by $O(t^{-a})$, and bound the remaining error by the sup-norm error divided by $a$. Taking $z_1, z_2$ to be two cluster points now yields limiting values $-|z_1 - z_2|^2$ and $|z_1 - z_2|^2$ for the same function $h$. Thus they coincide. A bounded trajectory with one cluster point converges, proving the proposition. ∎

**Corollary 11.1** (A universal threshold)**.** *Among positive damping parameters, the property that every regular cold trajectory converges for every finite-dimensional differentiable convex objective with globally Lipschitz gradient and nonempty minimizer set holds exactly for $\alpha \geq 3$. The same threshold holds when the objectives are restricted to globally $C^\infty$ convex functions with bounded derivatives of all positive orders. The failure already occurs in dimension two. For each $1 < \alpha < 3$ it can occur with a compact minimizer segment and a bounded trajectory; for $0 < \alpha \leq 1$ the example above has an unbounded minimizer set and an unbounded trajectory.*

**Proof.** The positive assertion follows from Proposition 11.2; it includes the smaller smooth class, since a bounded Hessian makes the gradient globally Lipschitz. For $1 < \alpha < 3$ apply Theorem 1. For $0 < \alpha \leq 1$, embed Proposition 11.1 in the plane by $F(s, y) = g(y)$ and start from $(0, 1)$ at rest. The tangential coordinate stays zero and the normal coordinate diverges. ∎

# 12 Scope of the construction

The final objective is static. The intermediate exact free normal tails are induction devices: later blocks intentionally destroy them to create the next steering episode. What

survives in the final orbit is each selected finite observation under a summable static force perturbation. Neither continuity of infinite-time parking endpoints nor uniform tangential velocity convergence through an elastic impact has been used.

The outer scale, inner core radius, finite certificate times, smoothing, and global completion are fixed before the physical height at each stage. The actual release proof supplies bounded normalized energy at a positive interior time before the penalty limit is applied. These two orderings avoid a circular choice in which lowering the height changes the normalized wall strength and invalidates a previously selected normal core.

The steering window closes at $\alpha = 3$: its two endpoints in (5.2) coincide, so small transition occupation and a diverging tangential clock are no longer obtained together. At $\alpha = 1$ the positive free-displacement parameter $a$ vanishes; the bounded construction does not pass to that endpoint. The low-damping example in Proposition 11.1 is separate and has neither a bounded orbit nor a compact minimizer set.

The function and all quantitative constants may depend on the fixed damping parameter. The construction gives neither one objective that works simultaneously for an interval of damping parameters nor a rate at which the observation times grow. It is an assertion about the continuous equation (1.1); it provides no discrete-time Nesterov counterexample.

# References


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